

Bachelor's Thesis (Academic Year 2026)

Adaptive Polarization Compensation for Optical Interconnects in Quantum Networks

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Abstract

Quantum networks rely on the transmission of photonic qubits over optical fiber links, where polarization drift caused by environmental and mechanical effects can significantly degrade measurement accuracy and entanglement fidelity. This thesis studies adaptive polarization compensation methods for optical quantum networking testbeds, focusing on automated calibration techniques using motorized polarization controllers and feedback from optical detectors. By evaluating the initial and final photon-count rates across repeated automated compensation runs, this work demonstrates the ability of the proposed method to recover a high-count polarization state and support more stable operation of fiber-based quantum communication systems.

Keywords

Adaptive polarization compensation, optical fiber, quantum networks, polarization drift, motorized polarization controller, single-photon detection, photon-count feedback.

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Chapter 1

Introduction

Scaling a single monolithic quantum processor to a large number of reliable qubits presents major physical and engineering challenges. In our current state, crosstalk, limited qubit connectivity, decoherence, wiring related complexities, and so on, make it difficult to continue to expand a single quantum processing unit [1, 2].

Distributed quantum computing provides an alternative approach in which several smaller quantum processing units are interconnected and operate as larger units [3, 4, 5]. This works by distributing computational tasks across multiple quantum processors instead of requiring every qubit to be contained within one device. Unique properties such as entanglement and quantum communication are then used to transfer quantum information or perform operations between physically separated nodes.

Photons are particularly suitable for interconnecting these processors because they can transmit quantum information through optical fiber [6, 3, 5]. They can also be routed using optical components such as switches, beam splitters, entangled photon-pair sources, and Bell-state analyzers [7, 8, 9]. Therefore, a network that interconnects several quantum processors can provide access to computational resources beyond those available within a single quantum device.

In the future, distributed quantum systems are expected to contain multiple nodes that dynamically share optical links and measurement [10, 11]. For instance, architectures such as Q-Fly use reconfigurable optical switches and shared Bell-state analyzers to establish connections between different pairs of quantum nodes [8]. Such architectures provide a possible path toward scalable quantum local-area networks and distributed quantum information processing systems.

However, establishing a route through a quantum network does not automatically create a usable physical link. The selected optical path must also be configured so that the transmitted photons arrive at the receiver under the required optical conditions such as polarization states [12, 13]. That said, as photons propagate through an optical fiber, their polarization state can be transformed by birefringence arising from bending, mechanical stress, temperature variation, imperfections within the fiber, and so on [14]. In this thesis, we will refer to this polarization change as *polarization drift*.

These polarization challenges become relevant in reconfigurable or deployed-fiber quantum networks, where different paths and changing environmental conditions can produce different optical transformations [15]. They can also reduce two-photon interference visibility, increase the quantum bit error rate, and decrease the fidelity of polarization-sensitive quantum operations. Hence, polarization compensation is an important operation for the scalability of distributed quantum systems.

This thesis investigates an adaptive polarization-compensation in which photon-count measurements from the detector are used as feedback for controlling a motorized fiber polarization controller. Different paddle-angle configurations are evaluated automatically, and the detector response is used to locate a configuration that produces a high photon-count rate. Although the experiment focuses on a single selected fiber link, it demonstrates a control process that may contribute to the automated configuration of future distributed quantum-information systems.

1.1 Motivation and Problem Statement

Polarization compensation on current quantum network testbeds is often performed manually or through partially automated procedures [13]. During manual compensation, an operator adjusts the polarization-controller paddles while observing the detector response. Although this approach is suitable for initial laboratory alignment, it is time-consuming, dependent on the operator, and difficult to reproduce consistently. These limitations become more significant as quantum systems expand beyond the laboratory [13, 16]. For instance, different optical paths may introduce different polarization transformations, and a setting obtained for one path may not remain usable after the network is reconfigured or the fiber conditions change [8, 15].

Therefore, an automated compensation system should evaluate the received signal, adjust the polarization controller, and locate a suitable configuration with minimal human intervention. In addition to automation, the compensation time must be reduced

so that optical links can be prepared without introducing excessive delay during network operation. The procedure should also be repeatable and capable of responding to changes in the measured detector output.

This process presents several practical challenges. One of these challenges is that photon-count measurements contain statistical fluctuations, and repeated measurements at the same controller setting may produce different results. Motorized paddles also require movement and settling time, which means that evaluating a large number of configurations can slow the compensation procedure. Moreover, the relationship between the paddle angles and the measured photon-count rate is not known analytically and must be determined experimentally.

Therefore, The compensation algorithm must balance two objectives: sufficiently exploring the paddle-angle space to locate a high-count configuration and limiting the number of physical movements and detector measurements required. A procedure that evaluates every possible configuration may provide broad coverage, but would be too slow for practical automated operation. On the other hand, a procedure that performs too few measurements may converge quickly but fails to locate an effective compensation setting.

1.2 Contributions

This thesis makes the following main contribution:

- An automated polarization-compensation system is developed for a fiber-optic quantum link. The system integrates a motorized polarization controller with single-photon detector feedback and uses a hybrid optimization procedure that combines a coarse two-dimensional search with quadratic response-surface refinement. Robust photon-count acquisition, including repeated sampling, median filtering, and invalid-measurement handling, is incorporated into the control procedure.

1.3 Thesis Organization

This thesis is organized as follows.

Chapter 2 introduces the background and theoretical foundations required to understand the work presented in this thesis. It begins with an overview of quantum computing, qubits, and photonic qubit encoding. The polarization states, Jones vectors, and birefringence are then discussed. The chapter also introduces quantum entanglement, quantum teleportation, the Hong–Ou–Mandel effect, and entanglement swapping. Finally, fiber-optic quantum networks and the polarization challenges associated with transmitting photons through optical fiber are presented.

Chapter 3 describes the experimental platform and the optical devices used in this work. Moreover, it introduces entangled photon-pair sources and the processes of spontaneous parametric down-conversion and spontaneous four-wave mixing. The chapter then describes single-mode fibers, polarization-maintaining fibers, waveplates, the motorized polarization controller, beam splitters, fiber couplers, and superconducting nanowire single-photon detectors.

Chapter 4 presents the experimental setup and the polarization-compensation system. It describes the complete optical signal path, the alignment of the motorized polarization controller, and the polarization-measurement configuration. The detector-channel configuration, photon-count integration, data-acquisition procedure, and software control architecture are also explored within this chapter. Finally, the objective function and the search and optimization procedure used by the polarization-compensation algorithm are described.

Chapter 5 presents the experimental results and evaluates the performance of the developed polarization-compensation system.

Chapter 6 concludes the thesis by summarizing the main findings and discussing the limitations of the present system.

Chapter 2

Background and Theoretical Foundations

2.1 Quantum Computing: An Overview

A quantum computer is a programmable computational device that encodes, processes, and retrieves information using controllable quantum states. Unlike classical computers, which operate on bits that take values of 0 or 1, quantum computers operate on quantum bits, or qubits, which can exist in superpositions of states [17, 18].

Let us take a coin toss as an example. If a coin is tossed into the air, the result is always head or tail, denoted by the classical results H or T equivalent to our classical bits 0 or 1. While the coin is flipping, we do not have a definitive answer to whether the outcome will be H or T, as the coin is undergoing continuous motion. Can this situation be considered an example of a superposition of states?

No.

A classical coin toss represents probabilistic uncertainty arising from incomplete knowledge of a system whose state is always well defined. Even when the coin is in motion, it occupies a definite physical configuration at every instant; the uncertainty lies only in the lack of information of the tosser.

This distinction is an important conceptual difference between classical and quantum systems. While classical bits represent definite logical values, quantum bits can exhibit non-classical behavior such as superposition and interference, which allow quantum

information to be processed in fundamentally different ways.

Superposition is a property of quantum systems in which a quantum state is expressed as a coherent linear combination of basis states. For a single qubit, the state may be written as

$$|\psi\rangle = \alpha |0\rangle + \beta |1\rangle, \quad (2.1)$$

where α and β are complex probability amplitudes that satisfy $|\alpha|^2 + |\beta|^2 = 1$. Measurement on a computational basis produces the outcome $|0\rangle$ with probability $|\alpha|^2$ and the outcome $|1\rangle$ with probability $|\beta|^2$. The relative phase between the amplitudes is also physically significant because it determines how the state behaves during quantum operations [18].

Quantum interference occurs when probability amplitudes of different quantum states combine. If two alternatives contribute amplitudes A_1 and A_2 to the same outcome, the probability of that result is

$$P = |A_1 + A_2|^2 = |A_1|^2 + |A_2|^2 + 2 \operatorname{Re}(A_1^* A_2). \quad (2.2)$$

The final term represents interference between the two alternatives. Depending on their relative phase, the amplitudes can interfere constructively or destructively. Quantum algorithms exploit this effect by amplifying amplitudes associated with desired outcomes while suppressing amplitudes associated with undesired outcomes [17, 18].

To describe these phenomena, qubits can be mathematically represented as vectors in a two or more dimensional complex Hilbert space [18]. The following section introduces this formal representation and discusses how qubit states can be visualized using the Bloch sphere.

2.2 The Qubit as a Physical Information Carrier

A qubit is a quantum bit that is analogous to the classical bits. However, while classical bits can take values of 0 and 1 (one of these states and only these states), a qubit can be represented mathematically, as a normalized vector in a two-dimensional complex Hilbert space:

$$|\psi\rangle = \alpha |0\rangle + \beta |1\rangle, \quad |\alpha|^2 + |\beta|^2 = 1. \quad (2.3)$$

In this representation, the qubit state $|\psi\rangle$ is not restricted to being in either $|0\rangle$ or $|1\rangle$; rather it can exist in a superposition of both. The coefficients α and β are complex probability amplitudes whose squared magnitudes determine the probabilities of obtaining the corresponding measurement outcomes. Specifically, a measurement in the computational basis yields outcome 0 with probability $|\alpha|^2$ and outcome 1 with probability $|\beta|^2$.

The basis states $|0\rangle$ and $|1\rangle$ play an analogous role to the classical bit values 0 and 1 as they show the two possible outcomes of a measurement in the computational basis. However, unlike classical bits, $|0\rangle$ and $|1\rangle$ are not numerical values but orthonormal basis vectors. Their meaning is therefore defined operationally by measurement rather than logical states.

Geometrically, the state of a qubit can be visualized on the Bloch sphere, where any pure state corresponds to a point on the surface of the sphere. Physical operations on a qubit correspond to rotations on this sphere. For instance, [fig. 2.1](#) illustrates some of the common qubit states:

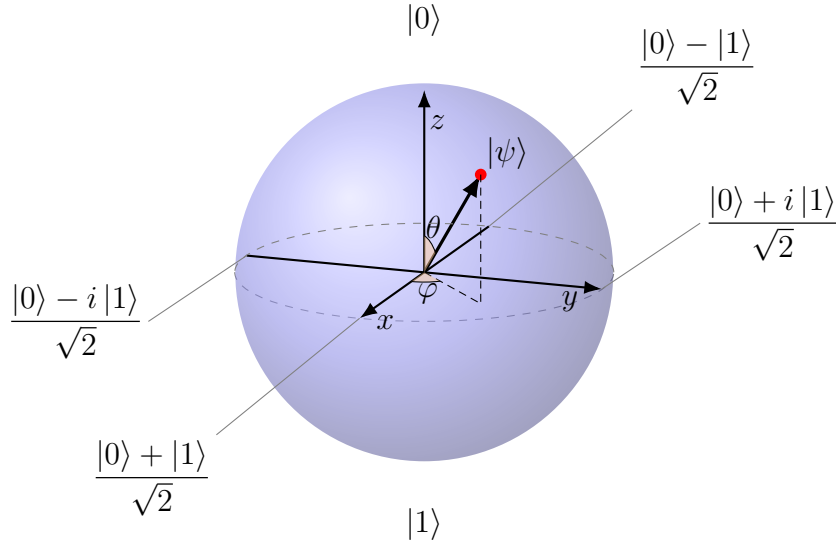


Figure 2.1: Examples of qubit states on the Bloch sphere.

Another distinguishing feature between qubits and classical bits is quantum entanglement. Entanglement is a phenomenon in which the quantum state of a composite system cannot be expressed as a product of the individual states of its constituent particles. Consequently, measurements performed on one particle can exhibit correlations with measurements performed on another, even when the particles are spatially separated [19]. Entanglement can occur in a system of two qubits, such as the Bell-state:

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle), \quad (2.4)$$

$$|\Phi^-\rangle = \frac{1}{\sqrt{2}}(|00\rangle - |11\rangle), \quad (2.5)$$

$$|\Psi^+\rangle = \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle), \quad (2.6)$$

$$|\Psi^-\rangle = \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle). \quad (2.7)$$

or in systems involving three or more qubits, such as the Greenberger–Horne–Zeilinger (GHZ) state:

$$|\text{GHZ}\rangle = \frac{1}{\sqrt{2}}(|000\rangle + |111\rangle). \quad (2.8)$$

Unlike classical correlations, entangled states cannot be expressed as products of individual qubit states, which shows a fundamental distinction between quantum and classical information.

2.3 Photonic Qubits and Polarization Encoding

2.3.1 Qubit Encoding

After discussing what qubits are and how they are represented, this section focuses on how physical qubits are realized and how information is encoded in physical systems.

One of the most important distinguishing features of a qubit is that it must possess two well-defined states, commonly denoted $|0\rangle$ and $|1\rangle$, which obey the rules of quantum amplitudes and probabilities. Moreover, the system must maintain phase coherence between $|0\rangle$ and $|1\rangle$, since loss of coherence destroys superposition and effectively reduces a qubit to a classical bit. In addition, the system must provide measurement capability that yields outcomes 0 or 1 with known probabilities. Given these requirements, an important question arises: what physical systems can be used to realize a qubit?

There are several physical implementations of qubits, and different research groups and

institutions are actively exploring these approaches. One prominent method is superconducting qubits, which are realized using small superconducting electrical circuits fabricated on a chip. These circuits behave as artificial atoms with two accessible energy levels corresponding to $|0\rangle$ and $|1\rangle$, and are typically controlled using microwave pulses at cryogenic temperatures [20, 21].

Another approach is to encode qubits using trapped ions, where the qubit is formed from the internal electronic states of an ion. Two common implementations include hyperfine qubits, which use two ground state hyperfine levels, and optical qubits, which use a ground state and an excited state. In both cases, qubit manipulation and measurement are performed using laser fields tuned to interact selectively with the relevant transitions. By detecting whether photons are emitted during these processes, the qubit state can be determined with very high fidelity [22]. Qubits can also be encoded using the spin states of electrons or nuclei, where the logical states correspond to different spin orientations [23, 24].

Photons are also another attractive platform for quantum information processing due to their weak interaction with the environment and their natural compatibility with optical fiber infrastructure. In photonic quantum systems, quantum information can be encoded using several degrees of freedom, including polarization, time-bin, frequency, and spatial modes [25, 26].

Due to the dual nature of light, exhibiting both wave-like and particle-like behavior, it is possible to exploit properties such as interference and polarization while simultaneously using the photon as the fundamental carrier of quantum information. This makes photonic qubits especially well suited for quantum communication and networking applications [25, 27]. As mentioned above, there are several ways to encode information using a qubit, but for the sake of this thesis, we will focus mainly on polarization encoding and how we can manipulate polarization states of photons in optical fiber links.

2.3.2 Polarization

Polarization describes the geometrical orientation of the oscillations of transverse waves. In the case of electromagnetic waves, these oscillations correspond to the electric and magnetic fields, which are always perpendicular to each other and to the direction of propagation. Since light behaves as an electromagnetic wave, polarization refers to the oscillatory behavior of these fields. In practice, however, polarization is almost always

defined using the electric field component. For example, consider the electromagnetic wave shown in [fig. 2.2](#), propagating through free space toward an observer. As illustrated, the electric field oscillates up and down while the wave propagates forward. This corresponds to a linearly polarized wave.

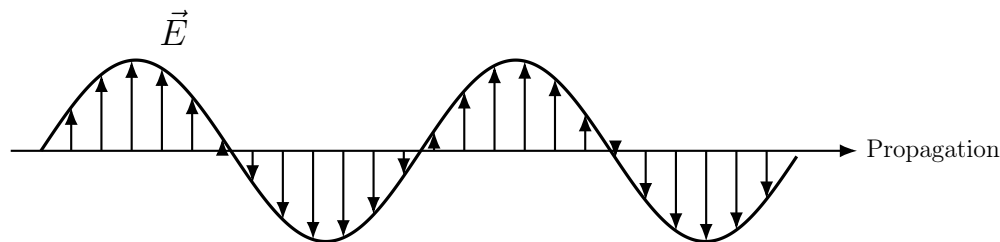


Figure 2.2: Visualization of a linearly polarized electromagnetic wave.

A polarized light wave is therefore a light wave in which the electric field oscillates in a specific, well-defined direction, rather than vibrating randomly in many directions. If we instead view the incoming wave head-on, as illustrated in [fig. 2.3](#), the electric field appears to oscillate along a fixed axis in the transverse plane.

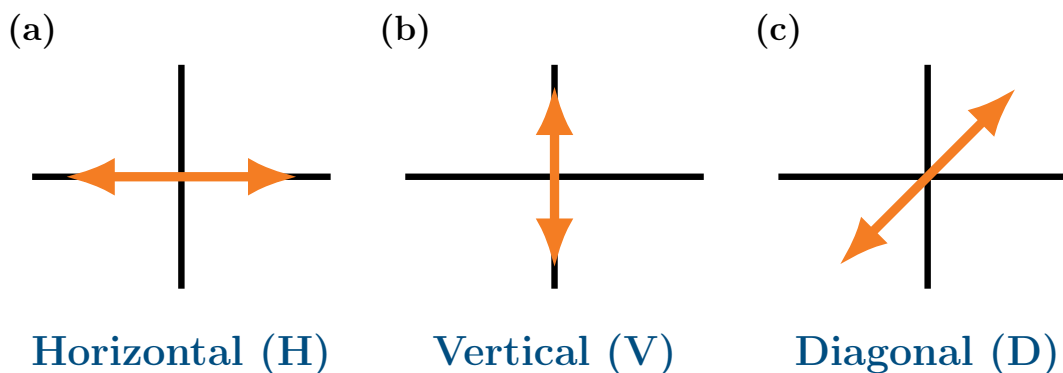


Figure 2.3: Examples of linearly polarized light: (a) horizontal polarization, (b) vertical polarization, and (c) diagonal polarization.

This form of polarization is known as linear polarization. The electric field may oscillate vertically, horizontally, or along any fixed diagonal direction. Other types of polarization also exist, such as elliptical and circular polarization, as shown in [fig. 2.4](#). In circular polarization, the electric field rotates with constant magnitude as the wave propagates, while in elliptical polarization the electric field traces an ellipse in the transverse plane.

Because polarization is often confused with other optical properties, it is important to emphasize that polarization is independent of intensity, frequency, or color. Moreover,

even though both the electric and magnetic fields participate in the propagation of light, polarization is conventionally described using the orientation and evolution of the electric field.

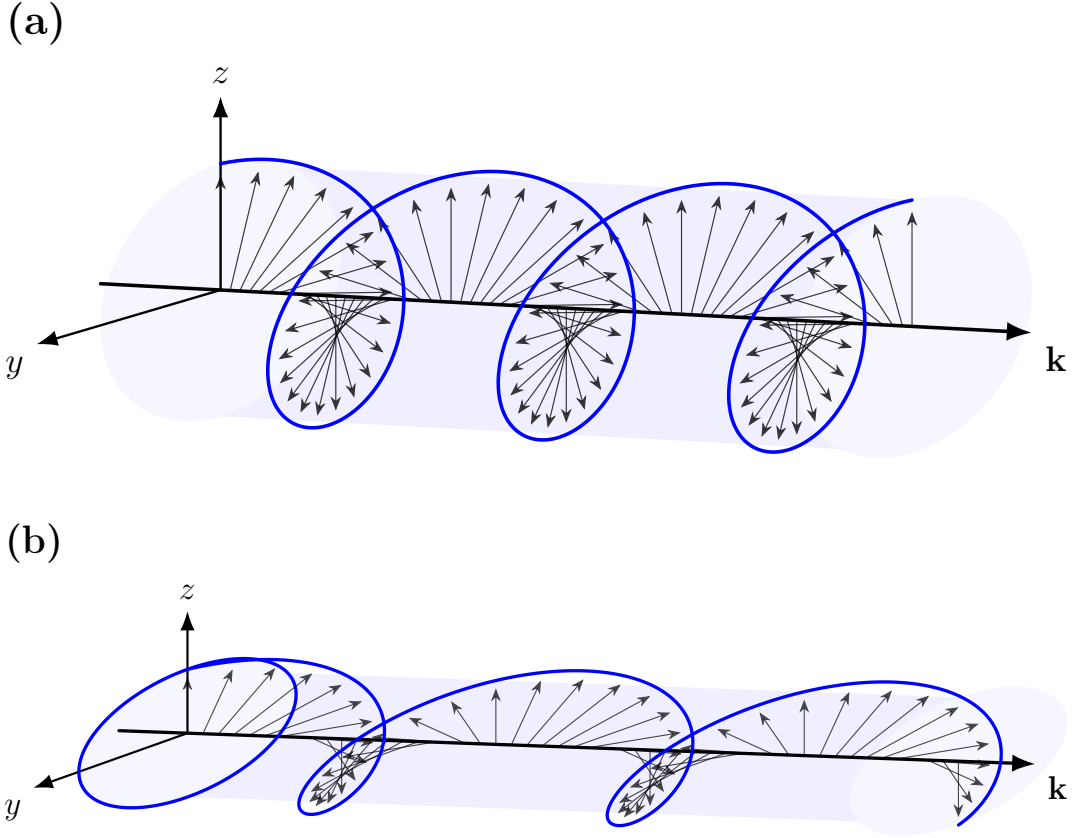


Figure 2.4: Examples of polarization states: (a) circularly polarized light and (b) elliptically polarized light.

2.3.3 Polarization Encoding

To encode information distinguishing between 0 and 1, we need two physical states that are orthogonal to each other (competently distinguishable from each other). One way of doing this is by mapping the horizontal polarization state and the vertical polarization state of a photon to represent the $|0\rangle$ and $|1\rangle$ states:

$$|0\rangle \equiv |H\rangle, \quad |1\rangle \equiv |V\rangle.$$

Another way of encoding information on a photon with the polarization degree of freedom is to use the diagonal basis:

$$|0\rangle \equiv |D\rangle, \quad |1\rangle \equiv |A\rangle.$$

$|D\rangle$ refers to the $+45^\circ$ and $|A\rangle$ refers to the -45° diagonals. In addition, we can also use the circular basis as the polarization states right-circular and left-circular to define the 0 and 1 states:

$$|0\rangle \equiv |R\rangle, \quad |1\rangle \equiv |L\rangle.$$

As mentioned above, since polarization can be a two-dimensional degree of freedom, a photon can exist in superposition of $|H\rangle$ and $|V\rangle$ states. Physically, this can correspond to

$$|\psi\rangle = \alpha|H\rangle + \beta|V\rangle \quad (\text{general linear}) \quad (2.9)$$

$$|+\rangle = \frac{|H\rangle + |V\rangle}{\sqrt{2}} \quad (\text{diagonal}), \quad |-\rangle = \frac{|H\rangle - |V\rangle}{\sqrt{2}} \quad (\text{anti-diagonal}) \quad (2.10)$$

$$|+i\rangle = \frac{|H\rangle + i|V\rangle}{\sqrt{2}} \quad (\text{right-circular}), \quad |-i\rangle = \frac{|H\rangle - i|V\rangle}{\sqrt{2}} \quad (\text{left-circular}) \quad (2.11)$$

2.4 Jones Vectors

As mentioned in [section 2.3.2](#), electromagnetic waves involve electric and magnetic fields that oscillate perpendicularly to each other and to the direction of the wave propagation. However, when describing polarization, it is sufficient to consider only the electric field component. The electric field $\vec{E}$ can therefore be written as:

$$\vec{E} = E_x \hat{e}_x + E_y \hat{e}_y + E_z \hat{e}_z,$$

or equivalently in matrix form as:

$$\vec{E} = \begin{bmatrix} E_x \\ E_y \\ E_z \end{bmatrix},$$

where E_x , E_y , and E_z are field amplitudes along orthogonal axes. Now let us assume that we have horizontally polarized light. In this case, the E_y and E_z components are 0, and the electric field reduces to:

$$\vec{E} = \begin{bmatrix} E_x \\ 0 \\ 0 \end{bmatrix}.$$

The remaining component E_x can be represented as a traveling wave:

$$E_x(t, z) = A \cos(\omega t - kz),$$

where t denotes time, z is the propagation direction, ω is the angular frequency, and k is the wave number. To generalize this traveling wave, we may include an additional phase term ϕ , which specifies the value of the wave at the origin:

$$E_x(t, z) = A \cos(\omega t - kz + \phi), \quad (2.12)$$

$$E_x(0, 0) = A \cos(\phi). \quad (2.13)$$

We call the parameter ϕ the *phase* of the wave. Changing the phase corresponds to a shift of the wave along its direction of propagation. For example, starting from a wave $\cos(\theta)$ and applying a phase shift of $-\frac{\pi}{2}$ yields a sine function:

$$\sin(\theta) = \cos\left(\theta - \frac{\pi}{2}\right). \quad (2.14)$$

Another useful relation for representing waves is Euler's formula:

$$e^{i\theta} = \cos \theta + i \sin \theta.$$

Before applying this expression, it is important to note that the physical electric field E_x is a real quantity. However, for mathematical convenience, we may represent it using a complex notation, where the real part corresponds to the physical field:

$$E_x = [\text{real part}] + i [\text{imaginary part}].$$

Using this representation, the electric field can be written as:

$$E_x = A \cos(\omega t - kz + \phi) + iA \sin(\omega t - kz + \phi).$$

Applying Euler's formula, this expression becomes

$$: E_x = A \cos(\omega t - kz + \phi) + iA \sin(\omega t - kz + \phi) \quad (2.15)$$

$$= Ae^{i(\omega t - kz + \phi)} \quad (2.16)$$

$$= Ae^{i(\omega t - kz) + i\phi} \quad (2.17)$$

$$= Ae^{i(\omega t - kz)} e^{i\phi}. \quad (2.18)$$

This form shows three distinct contributions: the amplitude, the phase, and the traveling wave factor. Following the same logic, we now consider the more general case in which both horizontal and vertical polarization components are present. The electric field can then be written as:

$$\vec{E} = \begin{bmatrix} A_x e^{i\phi_x} e^{i(\omega t - kz)} \\ A_y e^{i\phi_y} e^{i(\omega t - kz)} \\ 0 \end{bmatrix}. \quad (2.19)$$

Factoring out the common propagation term yields:

$$\vec{E} = \begin{bmatrix} A_x e^{i\phi_x} e^{i(\omega t - kz)} \\ A_y e^{i\phi_y} e^{i(\omega t - kz)} \\ 0 \end{bmatrix} = \begin{bmatrix} A_x e^{i\phi_x} \\ A_y e^{i\phi_y} \\ 0 \end{bmatrix} e^{i(\omega t - kz)}. \quad (2.20)$$

From this expression, we see that the polarization properties of the wave are fully determined by the relative amplitudes and phases contained in the equation:

$$\begin{bmatrix} A_x e^{i\phi_x} \\ A_y e^{i\phi_y} \\ 0 \end{bmatrix}. \quad (2.21)$$

The longitudinal component E_z can be ignored, since light is a transverse wave and the propagation direction is taken along the z -axis. We therefore arrive at the reduced two-dimensional representation:

$$\mathbf{J} \longrightarrow \begin{bmatrix} A_x e^{i\phi_x} \\ A_y e^{i\phi_y} \end{bmatrix}. \quad (2.22)$$

which is known as the *Jones vector*.

In fact, any polarization state of light can be expressed as a linear combination of horizontal and vertical polarization states, as illustrated in [eq. \(2.10\)](#) and [eq. \(2.11\)](#).

In Jones vector notation, this can be written as:

$$\mathbf{J} = A_x e^{i\phi_x} |H\rangle + A_y e^{i\phi_y} |V\rangle, \quad |H\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad |V\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

2.5 Birefringence

Materials can generally be classified into three groups based on their refractive indices in different directions [28]. Isotropic materials are materials that have a uniform refractive index in all directions $n_x = n_y = n_z$. Another type is uniaxial materials, which have one refractive index different from the other two, $n_x = n_y \neq n_z$. The last type is biaxial materials, which have three different refractive indices along the three principal directions, $n_x \neq n_y \neq n_z$.

Another important concept is the optic axis. [Figure 2.5](#) shows graphical representations of the refractive indices, which are known as index ellipsoids. For an isotropic material, the refractive index is the same in all directions, and the index surface is therefore a sphere. In a uniaxial material, the sphere is either stretched or compressed along one principal direction, producing an ellipsoid. The optical axis is the symmetry axis of this ellipsoid and represents the propagation direction along which the ordinary and extraordinary waves experience the same effective refractive index, meaning that no birefringence occurs along this direction.

For instance, if the index ellipsoid is compressed along its symmetry axis, as shown for the negative uniaxial crystal, the principal extraordinary refractive index is smaller than the ordinary refractive index. This index is known as n_e , which stands for the extraordinary refractive index. The two remaining equal indices are represented by n_o , which stands for the ordinary refractive index. The difference between these refractive indices is what we call birefringence:

$$\text{Birefringence:} \quad \Delta n = n_e - n_o$$

.

In this case, ordinary and extraordinary describe the two polarization modes of a birefringent crystal. Since the phase velocity is given by:

$$v_p = \frac{c}{n},$$

the component experiencing the larger refractive index propagates more slowly. In contrast, the component experiencing the smaller refractive index propagates faster and is known as the fast mode. Since $n_s > n_f$, the corresponding phase velocities satisfy:

$$v_s = \frac{c}{n_s} < \frac{c}{n_f} = v_f.$$

Returning to [fig. 2.5](#), if $n_e < n_o$, then the extraordinary component experiences the smaller refractive index and is therefore the fast mode, while the ordinary component is the slow mode. This type of material is known as a negative uniaxial crystal. On the other hand, if $n_e > n_o$, then the ordinary component is faster and is called the fast mode, while the extraordinary component is called the slow mode. This type of material is known as a positive uniaxial crystal.

Birefringence is therefore defined as the difference between the refractive indices experienced by the two orthogonal polarization modes. It may also be expressed in terms of the slow- and fast-axis refractive indices as:

$$|\Delta n| = n_s - n_f,$$

where n_s and n_f are the refractive indices associated with the slow and fast modes, respectively.

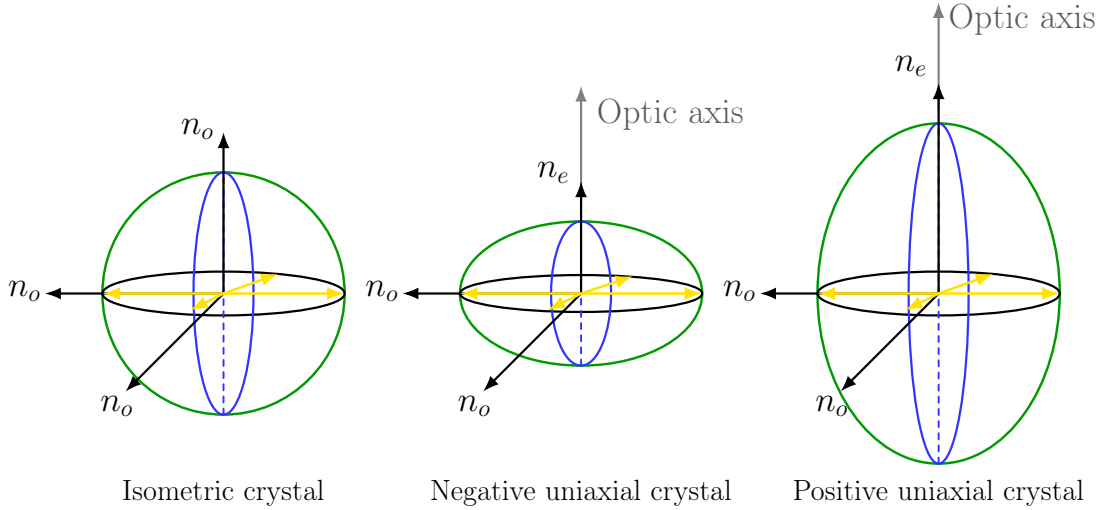


Figure 2.5: Refractive-index surfaces for isotropic, negative uniaxial, and positive uniaxial crystals.

Moreover, in a birefringent crystal, any arbitrary polarization state can be decomposed into two orthogonal polarization components corresponding to the ordinary and extraordinary modes of the crystal [29]. These components experience different refractive indices, n_o and n_e , and thus propagate with different phase velocities. Therefore, they accumulate a relative phase difference while propagating through the material. This polarization-dependent refractive index is important in nonlinear optical crystals, where it can be used to satisfy the phase-matching condition required for processes such as spontaneous parametric down-conversion, which will be described in the next chapter.

2.6 Quantum Entanglement

Now that we have seen how polarization states can be represented using Jones vectors and how photonic qubits may be encoded using polarization, we will turn our attention to systems involving multiple qubits. Although the polarization state of a single photon can be described independently, many quantum networking protocols rely on correlations between two or more photons. These correlations can give rise to a phenomenon known as quantum entanglement, which plays a very important role in quantum communication, Bell-state measurements, and entanglement swapping.

Unlike classical systems, where the properties of multiple objects can be described independently, entangled quantum systems must be described as a single composite system. In such systems, the state of one particle cannot be fully specified without

considering the state of the other particles, regardless of the physical distance between them [19]. However, since we have already introduced the different states possible because of entanglement, this section will try to answer the question "why do we care about entanglement?"

2.6.1 Quantum Teleportation

To explain quantum teleportation, we will use the example of Alice and Bob. Suppose that Alice wants to send an unknown qubit to Bob. Now, the obvious solution would be to send it classically. However, this is impossible because the state cannot be measured without collapsing the quantum state. However, if Alice and Bob share an entangled pair, Alice can perform a Bell-state measurement and send only two classical bits to Bob. A Bell-state measurement is a quantum measurement performed on a pair of qubits that determines which of the four Bell-states, represented in [eq. \(2.10\)](#) and [eq. \(2.11\)](#), the qubits occupy. Bob, in turn, can apply a correction operation and reconstruct the original quantum state [30]. In this process, the original state held by Alice is destroyed, which shows that quantum teleportation does not violate the no-cloning theorem [31]. For instance, let us assume that Alice has the following state:

$$|\psi\rangle_1 = \alpha|0\rangle_1 + \beta|1\rangle_1$$

where

$$|\alpha|^2 + |\beta|^2 = 1.$$

Alice and Bob also share the Bell-state:

$$|\Phi^+\rangle_{23} = \frac{1}{\sqrt{2}} (|00\rangle_{23} + |11\rangle_{23})$$

Qubit 2 belongs to Alice, and qubit 3 belongs to Bob. The combined initial state then becomes a tensor product between Alice's unknown qubit and the shared entangled state:

$$|\Psi\rangle_{123} = |\psi\rangle_1 \otimes |\Phi^+\rangle_{23}$$

We should note that the tensor product does not imply entanglement; it only refers to placing the systems together.

Next, by substituting the states, we obtain:

$$|\Psi\rangle_{123} = (\alpha|0\rangle_1 + \beta|1\rangle_1) \otimes \frac{1}{\sqrt{2}} (|00\rangle_{23} + |11\rangle_{23}) \quad (2.23)$$

$$= \frac{1}{\sqrt{2}} (\alpha|000\rangle + \alpha|011\rangle + \beta|100\rangle + \beta|111\rangle). \quad (2.24)$$

Up to this point, we have not yet begun the actual teleportation process. The process begins with Alice performing a Bell-state measurement on qubits 1 and 2. The four possible states are:

$$|\Phi^+\rangle_{12} = \frac{1}{\sqrt{2}} (|00\rangle_{12} + |11\rangle_{12})$$

$$|\Phi^-\rangle_{12} = \frac{1}{\sqrt{2}} (|00\rangle_{12} - |11\rangle_{12})$$

$$|\Psi^+\rangle_{12} = \frac{1}{\sqrt{2}} (|01\rangle_{12} + |10\rangle_{12})$$

$$|\Psi^-\rangle_{12} = \frac{1}{\sqrt{2}} (|01\rangle_{12} - |10\rangle_{12}).$$

This can be rewritten as:

$$|00\rangle_{12} = \frac{1}{\sqrt{2}} (|\Phi^+\rangle_{12} + |\Phi^-\rangle_{12})$$

$$|11\rangle_{12} = \frac{1}{\sqrt{2}} (|\Phi^+\rangle_{12} - |\Phi^-\rangle_{12})$$

$$|01\rangle_{12} = \frac{1}{\sqrt{2}} (|\Psi^+\rangle_{12} + |\Psi^-\rangle_{12})$$

$$|10\rangle_{12} = \frac{1}{\sqrt{2}} (|\Psi^+\rangle_{12} - |\Psi^-\rangle_{12}).$$

Next, we group the three-qubit state, [eq. \(2.23\)](#), according to Alice's two qubits and write each term as "Alice's pair" $\otimes$ "Bob's qubit":

$$|\Psi\rangle_{123} = \frac{1}{\sqrt{2}} (\alpha|00\rangle_{12}|0\rangle_3 + \alpha|01\rangle_{12}|1\rangle_3 + \beta|10\rangle_{12}|0\rangle_3 + \beta|11\rangle_{12}|1\rangle_3).$$

Finally, we substitute the Bell-basis identities discussed above and, after collecting the terms, we obtain:

$$\begin{aligned} |\Psi\rangle_{123} = \frac{1}{2} & \left[|\Phi^+\rangle_{12} (\alpha|0\rangle_3 + \beta|1\rangle_3) + |\Phi^-\rangle_{12} (\alpha|0\rangle_3 - \beta|1\rangle_3) \right. \\ & \left. + |\Psi^+\rangle_{12} (\beta|0\rangle_3 + \alpha|1\rangle_3) + |\Psi^-\rangle_{12} (-\beta|0\rangle_3 + \alpha|1\rangle_3) \right]. \end{aligned} \quad (2.25)$$

[Equation \(2.25\)](#) shows that Alice's measurement result determines which version of the original state Bob possesses. Thus, if Alice measures $|\Phi^+\rangle_{12}$, Bob already possesses the original state $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$, which means that no correction is required. If Alice measures $|\Phi^-\rangle_{12}$, then Bob has $|\psi\rangle = \alpha|0\rangle - \beta|1\rangle$, which is the original state with a Z error, so Bob applies the Pauli- Z gate. Similarly, if Alice measures $|\Psi^+\rangle_{12}$ or $|\Psi^-\rangle_{12}$, then Bob applies the Pauli- X and XZ operators to perform the appropriate correction and reconstruct the original state.

2.6.2 Entanglement Swapping

Before we introduce Entanglement Swapping as a topic, it is important to make a distinction between Quantum Teleportation, described in [section 2.6.1](#), and entanglement swapping. In teleportation, we begin with an unknown qubit and a shared bell pair to end with the unknown transferred qubit. The goal is simply to move a quantum state. However, in entanglement swapping, no unknown state is transmitted. Instead, two pairs of local or short-range entangled pairs are consumed to establish entanglement between two qubits that were not previously entangled [\[32\]](#). For instance, let us assume that we have a four-qubit system. Initially we have the following states where qubit 1 is entangled with qubit 4 and qubit 2 is entangled with qubit 3:

$$|\Phi^+\rangle_{12} = \frac{1}{\sqrt{2}} (|00\rangle + |11\rangle) \quad |\Phi^+\rangle_{34} = \frac{1}{\sqrt{2}} (|00\rangle + |11\rangle)$$

The total state is given by:

$$|\Psi\rangle = |\Phi^+\rangle_{12} \otimes |\Phi^+\rangle_{34},$$

which becomes the following when expanded:

$$|\Psi\rangle = \frac{1}{2} (|0000\rangle + |0011\rangle + |1100\rangle + |1111\rangle). \quad (2.26)$$

Next, the middle node performs a Bell-state measurement on qubits 2 and 3. Therefore, we want all terms to be rearranged as $(23) \otimes (14)$. From [eq. \(2.26\)](#), the first term is:

$$|0000\rangle = |0\rangle_1 |0\rangle_2 |0\rangle_3 |0\rangle_4$$

Thus, we group 23 and 14:

$$= |00\rangle_{23} |00\rangle_{14}.$$

We follow the same pattern for the second, third and fourth term:

$$|0011\rangle = |0\rangle_1 |0\rangle_2 |1\rangle_3 |1\rangle_4$$

$$= |01\rangle_{23} |01\rangle_{14}.$$

$$|1100\rangle = |1\rangle_1 |1\rangle_2 |0\rangle_3 |0\rangle_4$$

$$= |10\rangle_{23} |10\rangle_{14}.$$

$$|1111\rangle = |1\rangle_1|1\rangle_2|1\rangle_3|1\rangle_4$$

$$= |11\rangle_{23}|11\rangle_{14}.$$

Substituting into [eq. \(2.26\)](#), we get:

$$|\Psi\rangle = \frac{1}{2} \left(|00\rangle_{23}|00\rangle_{14} + |01\rangle_{23}|01\rangle_{14} + |10\rangle_{23}|10\rangle_{14} + |11\rangle_{23}|11\rangle_{14} \right). \quad (2.27)$$

The next step is to rewrite the computational basis as did for the teleportation:

$$|00\rangle = \frac{1}{\sqrt{2}} (|\Phi^+\rangle + |\Phi^-\rangle) \quad (2.28)$$

$$|11\rangle = \frac{1}{\sqrt{2}} (|\Phi^+\rangle - |\Phi^-\rangle) \quad (2.29)$$

$$|01\rangle = \frac{1}{\sqrt{2}} (|\Psi^+\rangle + |\Psi^-\rangle) \quad (2.30)$$

$$|10\rangle = \frac{1}{\sqrt{2}} (|\Psi^+\rangle - |\Psi^-\rangle). \quad (2.31)$$

Next, we substitute the above terms in [eq. \(2.27\)](#):

$$\begin{aligned} |\Psi\rangle &= \frac{1}{2} \left(|00\rangle_{23}|00\rangle_{14} + |01\rangle_{23}|01\rangle_{14} + |10\rangle_{23}|10\rangle_{14} + |11\rangle_{23}|11\rangle_{14} \right) \\ &= \frac{1}{4} \left[\left(|\Phi^+\rangle_{23}|\Phi^+\rangle_{14} + |\Phi^+\rangle_{23}|\Phi^-\rangle_{14} + |\Phi^-\rangle_{23}|\Phi^+\rangle_{14} + |\Phi^-\rangle_{23}|\Phi^-\rangle_{14} \right) \right. \\ &\quad + \left(|\Psi^+\rangle_{23}|\Psi^+\rangle_{14} + |\Psi^+\rangle_{23}|\Psi^-\rangle_{14} + |\Psi^-\rangle_{23}|\Psi^+\rangle_{14} + |\Psi^-\rangle_{23}|\Psi^-\rangle_{14} \right) \\ &\quad + \left(|\Psi^+\rangle_{23}|\Psi^+\rangle_{14} - |\Psi^+\rangle_{23}|\Psi^-\rangle_{14} - |\Psi^-\rangle_{23}|\Psi^+\rangle_{14} + |\Psi^-\rangle_{23}|\Psi^-\rangle_{14} \right) \\ &\quad \left. + \left(|\Phi^+\rangle_{23}|\Phi^+\rangle_{14} - |\Phi^+\rangle_{23}|\Phi^-\rangle_{14} - |\Phi^-\rangle_{23}|\Phi^+\rangle_{14} + |\Phi^-\rangle_{23}|\Phi^-\rangle_{14} \right) \right]. \end{aligned}$$

Finally, after canceling like terms and rearranging, we are left with:

$$\begin{aligned} |\Phi^+\rangle_{12} \otimes |\Phi^+\rangle_{34} = \frac{1}{2} & \left(|\Phi^+\rangle_{23} |\Phi^+\rangle_{14} + |\Phi^-\rangle_{23} |\Phi^-\rangle_{14} \right. \\ & \left. + |\Psi^+\rangle_{23} |\Psi^+\rangle_{14} + |\Psi^-\rangle_{23} |\Psi^-\rangle_{14} \right). \end{aligned} \quad (2.32)$$

Now, if the Bell-state measurement on qubits 2 and 3 gives $|\Psi^+\rangle_{23}$, the state will collapse into $|\Psi^+\rangle_{14}$, thus giving us an entanglement between qubit 1 and qubit 4. Note that this happened even though the qubit 1 never interacted with the qubit 4. This phenomenon is known as *Entanglement Swapping*.

2.6.3 Hong-Ou-Mandel Effect

In the previous section, we introduced Bell-state measurements as a fundamental component of quantum teleportation. In photonic quantum communication systems, Bell-state measurements are usually implemented using beam splitters and two-photon interference. One of the most important interference effects that enables this operation is the Hong-Ou-Mandel (HOM) effect.

To understand HOM, let us first consider two identical photons entering a lossless 50:50 beam splitter from opposite sides at the same time and exiting at the same time as well. Each photon has two options:

1. Transmit through the beam splitter.
2. Reflect from the beam splitter.

Therefore, after the two photons pass through the beam splitter, the four possible outcomes are:

As shown in [fig. 2.6](#), (a) and (b) result in the detection of one photon at one output and the other at a different output. This means that we have a coincidence click, where both detectors at the end of our paths click at the same time. If the photons are distinguishable, such that one is in the H polarization state and the other is in the V polarization state, then the detector behavior is simply random, and we obtain coincidence clicks at the detectors attached to the ends of the photon paths. However, if the photons are truly indistinguishable such that each photon occupies the same

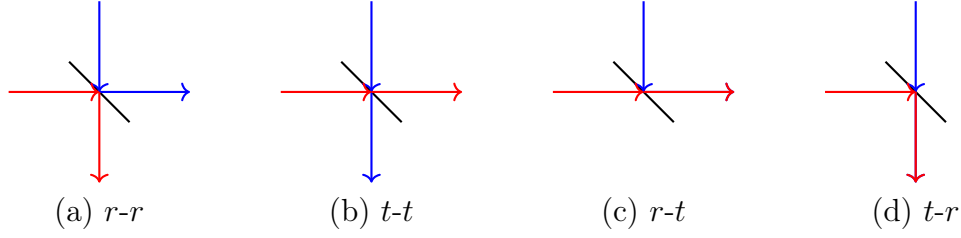


Figure 2.6: Possible paths for two photons incident on a 50:50 beam splitter. The blue and red arrows represent the paths of the two photons. (a) Both photons are reflected ($r-r$), (b) both photons are transmitted ($t-t$), (c) reflection–transmission ($r-t$), and (d) transmission–reflection ($t-r$).

polarization, frequency and wavelength, temporal wave-packet, arrival time at the beam splitter and so on, then according to quantum mechanics the two alternatives that produce one photon in each output (cases (a) and (b)) never occur ($P_{\text{coincidence}} = 0$). This phenomenon is known as the HOM effect.

2.7 Fiber-Optic Quantum Networks and the Polarization Challenge

Although early quantum computing research focused on isolated quantum processors, many practical applications require multiple quantum systems to communicate and operate together [33]. Moreover, scaling a single monolithic quantum processor introduces physical and engineering challenges, including decoherence, crosstalk, limited connectivity, control complexity, and constraints on wiring and processor topology. These limitations have motivated the development of distributed quantum systems in which separate quantum processors are connected through quantum communication links [1, 33].

A quantum network enables quantum information and entanglement to be distributed between spatially separated nodes. Such networks may contain quantum processors, quantum memories, quantum transducers, switches, photon sources, and Bell-state analyzers. Operations such as quantum teleportation and entanglement swapping, introduced in previous sections, allow quantum states and entanglement to be extended across these distributed systems [34].

Even when stationary qubits inside the network nodes are implemented using superconducting circuits, trapped ions, atomic systems, or other technologies, photons are commonly used as flying qubits to transmit quantum information between the nodes.

This is because photons interact weakly with the environment and can propagate over long distances through the existing optical communication infrastructure [25]. Therefore, optical fiber is one of the most important physical channels for quantum communication.

However, transmitting polarization-encoded photonic qubits through optical fiber introduces its own challenge. Theoretically, the horizontal and vertical polarization components would propagate in a predictable manner. However, in reality, manufacturing imperfections, mechanical stress, bending, vibration, and temperature fluctuations introduce birefringence into the fiber. As discussed in Section 2.5, birefringence causes orthogonal polarization components to experience different refractive indices and accumulate a relative phase difference. The polarization transformation introduced by the fiber can be represented using a Jones matrix. If the transmitted polarization state is represented by the Jones vector $\vec{J}_{\text{in}}$, the received state may be written as:

$$\vec{J}_{\text{received}}(t) = \mathbf{J}_{\text{fiber}}(t)\vec{J}_{\text{in}}, \quad (2.33)$$

where $\mathbf{J}_{\text{fiber}}(t)$ represents the fiber polarization transformation. The explicit time dependence is included to show that this transformation may change as the environmental and mechanical conditions surrounding the fiber also change.

As a result, the polarization state measured at the receiver may differ from the state prepared at the transmitter. For polarization-encoded quantum communication, this misalignment can reduce measurement accuracy, decrease interference visibility, increase the quantum bit error rate, and degrade the fidelity of entangled states. It may therefore affect fundamental operations such as Bell-state measurements, quantum teleportation, and entanglement swapping.

The central problem addressed in this thesis is how to compensate for this unknown and time-varying polarization transformation. A polarization controller can apply an additional controllable transformation, represented by $\mathbf{J}_C(\theta)$, where θ denotes the controller settings. The resulting output state will then be:

$$\vec{J}_{\text{out}}(t) = \mathbf{J}_C(\theta)\mathbf{J}_{\text{fiber}}(t)\vec{J}_{\text{in}}. \quad (2.34)$$

The objective is to determine controller settings that counteract the fiber-induced transformation and restore the desired polarization state at the receiver. Since the fiber transformation changes over time, a fixed correction may not remain effective. This

motivates an adaptive compensation approach in which detector measurements are used as feedback to automatically adjust the polarization controller.

The following chapter introduces the experimental platform and optical components used to investigate this problem, including the photon-pair source, optical fibers, polarization-control devices, beam splitters, fiber couplers, and single-photon detectors. Using this theoretical foundation, the next chapter describes how these components are integrated to form the experimental polarization-compensation system.

Chapter 3

Experimental Platform and Optical Devices

The previous chapter introduced the theoretical role of polarization in photonic quantum information systems and discussed how polarization drift in optical fibers can degrade quantum communication protocols. In order to study this problem experimentally, this chapter introduces the laboratory platform and optical devices used in this work.

3.1 Overview of the Experimental Platform

The experimental platform consists of three main stages: entangled-photon generation, photon distribution, and photon detection. First, the Entangled Photon Pair Source (EPPS), described in detail [section 3.2](#), generates pairs of photons. The photons are then transmitted through optical fibers to the switching layer, which establishes the optical paths required to connect the selected EPPS nodes to a Bell-State Analyzer (BSA). The BSA performs joint measurements on photons received from separate nodes, enabling operations such as entanglement swapping.

In the current experimental configuration, the Motorized Polarization Controller (MPC) is positioned after the switching layer and before the Fiber Polarizing Beam Splitter (FPBS) and Superconducting Nanowire Single-Photon Detectors (SNSPDs). This position allows the MPC to compensate for the polarization changes accumulated as the photons propagate through the optical fibers and switching components. After polarization compensation, the FPBS separates the incoming photons according to their

polarization states, and the resulting photons are detected by the SNSPDs. The overall optical path utilized in this experiment is discussed in depth in [section 4.1](#).

3.2 Entangled Photon Pair (EPPS)

In [section 2.3.3](#) we saw the theoretical representation of states $|0\rangle$ and $|1\rangle$ using polarization encoding. In this section, we will discuss how this method is completed physically and the two main ways of generating entangled photon pairs, Spontaneous Parametric Down-Conversion (SPDC) and Spontaneous Four-Wave Mixing.

3.2.1 Spontaneous Parametric Down-Conversion

Spontaneous Parametric Down-Conversion (SPDC) is a process that occurs when a high-energy pump photon enters a nonlinear material and is converted into two lower-energy photons called the signal and idler photons. In this process, both energy and momentum are conserved. For instance, as shown in [fig. 3.1](#), the pump photon has a higher energy E_p , and hence the two generated photons share that energy as:

$$E_p = E_s + E_i \quad (3.1)$$

In addition, SPDC also has to satisfy momentum conservation, or what is usually referred to as phase matching, as:

$$\vec{k}_p = \vec{k}_s + \vec{k}_i, \quad (3.2)$$

where $\vec{k}_p$, $\vec{k}_s$, and $\vec{k}_i$ are the wave vectors of the pump, signal, and idler photons, respectively.

In everyday optics, most materials behave linearly. Hence, photons that pass through a linear lens or glass block may reflect, refract, or scatter; however, they cannot spontaneously split into new frequencies. To create signal and idler photons, the material must allow interactions between multiple optical fields. Therefore, only certain materials possessing strong nonlinear optical properties can support the SPDC process.

The nonlinear optical properties required for SPDC originate from the interaction be-

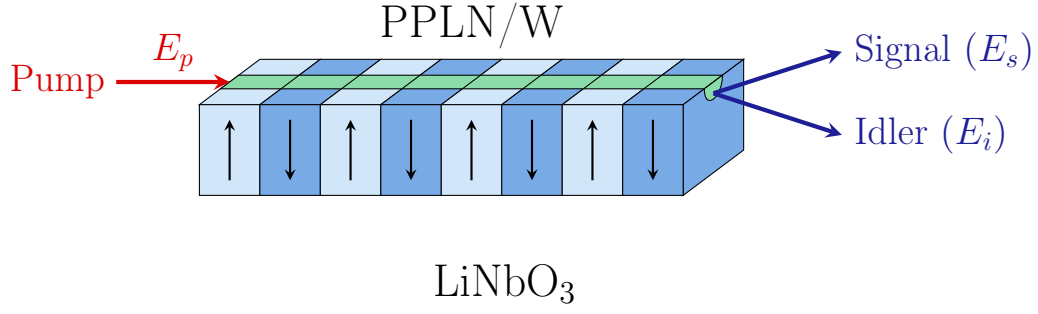


Figure 3.1: Spontaneous parametric down-conversion in a nonlinear crystal.

tween the electromagnetic field and the electrons within the crystal. Unlike linear optical materials, nonlinear materials exhibit higher-order responses to an applied electric field. For instance, SPDC is a second-order nonlinear optical process due to the material's $\chi^{(2)}$ susceptibility, which enables the conversion of a single pump photon into a pair of lower-energy photons while satisfying both energy and momentum conservation.

An example of a non linear crystal $\chi^{(2)}$ is periodically poled lithium niobate (PPLN), chemically represented as LiNbO_3 and shown in [fig. 3.2](#). It is called periodically poled because the orientation of the crystal's ferroelectric domains is reversed at regular spatial intervals. These repeated domain reversals cause the sign of the second-order nonlinear coefficient $\chi^{(2)}$ to alternate along the crystal. This, in turn, allows PPLN to achieve quasi-phase matching instead of relying on the natural birefringence of the crystal.

Quasi-phase matching is a method utilized in nonlinear crystals to prevent the photon-pair amplitudes generated at different positions inside the crystal from canceling each other out. For instance, in the SPDC process, one pump photon produces a signal photon and an idler photon:

$$\omega_p = \omega_s + \omega_i. \quad (3.3)$$

However, energy conservation alone is not enough. For perfect phase matching, the interacting waves must also maintain the correct phase relationship, such that $\Delta k = 0$ in $\Delta k = k_p - k_s - k_i$. If $\Delta k \neq 0$, the photon-pair probability amplitudes generated near the beginning of the crystal gradually move out of phase with those generated farther along the crystal. After some propagation distance, the newly generated amplitudes begin to cancel the previously generated ones.

To avoid this, in PPLN, the orientation of the lithium-niobate ferroelectric domains is

reversed periodically as:

$$+\chi^{(2)}, -\chi^{(2)}, +\chi^{(2)}, -\chi^{(2)}, \dots \quad (3.4)$$

so that just before the generated amplitudes begin to interfere destructively, the reversal introduces an effective additional phase shift of π , which causes the newly generated amplitudes to line up constructively. Other nonlinear optical materials, such as beta barium borate (BBO), can also be used for SPDC. However, the discussion in this work focuses on PPLN.

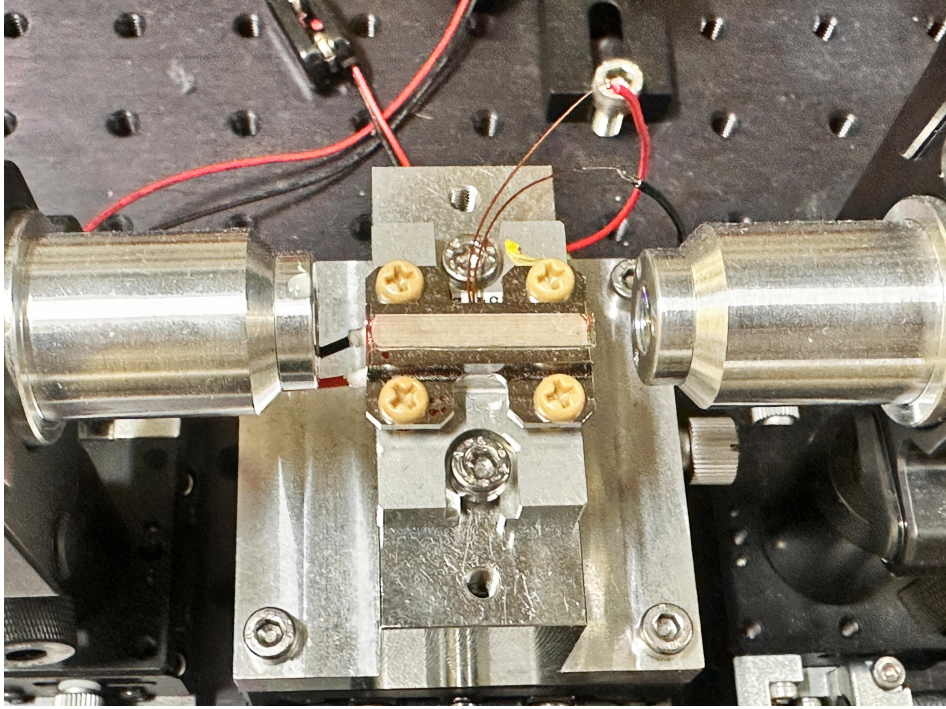


Figure 3.2: Periodically poled lithium niobate (PPLN) crystal.

Depending on the chosen phase-matching configuration, different SPDC configurations can be created. These configurations are classified as Type-0, Type-I, and Type-II. Type-0 SPDC occurs when the pump, signal, and idler all possess the same polarization. Polarization in this context refers to the polarization of the pump relative to the crystal axes. This could be in the forms of $e \rightarrow e + e$ or $o \rightarrow o + o$, where e and o represent extraordinary and ordinary polarization modes.

However, unlike Type-0 SPDC, Type-I SPDC occurs when the generated signal and idler photons share the same polarization mode, but this mode differs from that of the pump photon. This can be represented as $|e\rangle_p \rightarrow |o\rangle_s |o\rangle_i$ or $|o\rangle_p \rightarrow |e\rangle_s |e\rangle_i$. Moreover, the signal and idler photons propagate in a polarization mode orthogonal to the pump

photon's polarization mode.

However, in Type-II SPDC, the generated signal and idler photons emerge with orthogonal polarization states. For instance, this can be represented as $|e\rangle_p \rightarrow |e\rangle_s|o\rangle_i$ or $|e\rangle_p \rightarrow |o\rangle_s|e\rangle_i$. Since there are two distinguishable polarization modes, these states can be represented as $|HV\rangle$ or $|VH\rangle$.

3.2.2 Spontaneous Four-Wave Mixing

In the previous section, we saw that SPDC originates from the second-order nonlinear susceptibility $\chi^{(2)}$ of a material. However, Spontaneous Four-Wave Mixing (SFWM) arises from the third-order nonlinear susceptibility $\chi^{(3)}$, which means that we have four optical fields interacting. To understand this difference, let us investigate the physical meaning of nonlinearity order.

Polarization can be expanded as a power series, with the nonlinearity order as:

$$P = \epsilon_0 (\chi^{(1)}E + \chi^{(2)}E^2 + \chi^{(3)}E^3 + \dots) \quad (3.5)$$

where P is the induced electric polarization, ϵ_0 is the vacuum permittivity, E is the applied electric field, and $\chi^{(n)}$ denotes the n th-order electric susceptibility of the material. The first-order term, $P = \epsilon_0\chi^{(1)}E$, refers to linear optics, which we described as linear transformations, refraction, reflection, and birefringence in the previous section. In this process, the output light has the same frequency as the input light. The second-order nonlinearity is the process utilized in SPDC. As we can see in [eq. \(3.5\)](#), the nonlinear medium of $\chi^{(2)}$ depends on E^2 , hence:

$$\begin{aligned} E &= E_0 \cos(\omega t) \\ E^2 &= (E_0 \cos(\omega t))^2 \\ E^2 &= E_0^2 \cos^2(\omega t) \end{aligned}$$

Now using the double-angle identity:

$$\cos(2x) = 2 \cos^2(x) - 1 \quad (3.6)$$

$$\cos^2(x) = \frac{1 + \cos(2x)}{2}. \quad (3.7)$$

Replacing the variable x with ωt , we obtain the following:

$$\cos^2(\omega t) = \frac{1 + \cos(2\omega t)}{2}. \quad (3.8)$$

Substituting back into our earlier equation:

$$E^2 = E_0^2 \left(\frac{1 + \cos(2\omega t)}{2} \right) \quad (3.9)$$

$$= \frac{E_0^2}{2} + \frac{E_0^2}{2} \cos(2\omega t). \quad (3.10)$$

From [eq. \(3.10\)](#), we can see that the nonlinear response now contains a term oscillating at 2ω , despite the initial field containing ω . This is also the origin of second-harmonic generation and other three-wave mixing processes such as SPDC. Likewise, for four-wave mixing, we follow the same approach and get:

$$E^3 = \frac{3E_0^3}{4} \cos(\omega t) + \frac{E_0^3}{4} \cos(3\omega t). \quad (3.11)$$

Similarly, we can see that we now have a 3ω term, despite the initial field containing ω . This process is called four-wave mixing.

3.2.3 Overview of the EPPS Architecture

A detailed discussion of the internal design and construction of an Entangled Photon Pair Source (EPPS) is outside the scope of this thesis. However, its general architecture can be described using several main functional stages. First, a pulsed laser

provides the optical pump signal. In some EPPS implementations, the pump wavelength is converted through second-harmonic generation (SHG) before being directed into a nonlinear crystal. Inside the nonlinear crystal, spontaneous parametric down-conversion (SPDC) probabilistically converts a pump photon into two lower-energy photons, commonly referred to as the signal and idler photons.

The generated photons are then separated into two optical output paths according to their wavelength, polarization, or spatial mode. Each path may contain filtering, coupling, and polarization-control components before being connected to an optical fiber. The two photons can then be distributed to separate quantum-network nodes, detectors, or a Bell-state analyzer. From a network perspective, the EPPS can therefore be represented as a source node with one optical pump input and two quantum optical outputs carrying an entangled photon pair. [Figure 3.3](#) presents an example of the optical hardware setup of the EPPS.

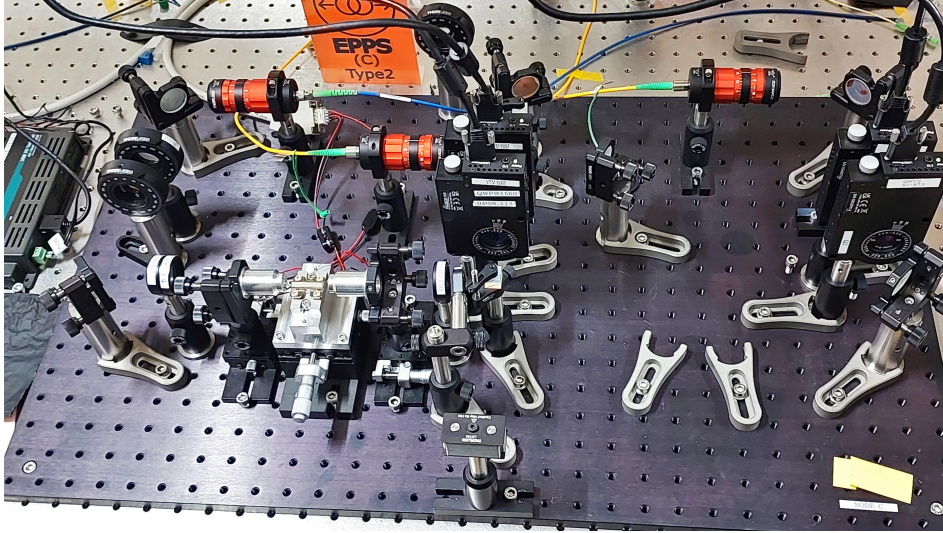


Figure 3.3: General architecture of an Entangled Photon Pair Source (EPPS)

3.3 Optical Waveguides and Fiber Links

Optical communication systems require structures that confine and guide light between different components. These structures are generally called optical waveguides. This section first introduces the basic concepts of optical waveguides before discussing the optical fibers used as communication channels in quantum networks.

3.3.1 Optical Waveguides

An optical waveguide is a dielectric structure that confines light in one or both transverse directions while allowing it to propagate along a longitudinal direction. A common dielectric optical waveguide consists of a guiding region with refractive index n_1 surrounded by material with a lower refractive index n_2 , where $n_1 > n_2$. This refractive-index contrast confines the supported optical modes to the guiding region [14].

The permitted field distributions inside a waveguide are referred to as guided modes. Depending on its dimensions, refractive-index profile, and operating wavelength, a waveguide may support a single spatial mode or multiple spatial modes. Single-mode propagation is important in quantum optical systems because it reduces spatial-mode mismatch between photons and improves their coupling and interference.

Optical waveguides can be constructed in several geometries. Planar and strip waveguides are commonly used in integrated photonic devices, while optical fibers are cylindrical waveguides designed to transmit light over longer distances. In the experimental platform used in this work, the PPLN waveguide confines the interacting optical fields within the nonlinear crystal, whereas the optical fiber carries the generated photons between the different components of the quantum link.

3.3.2 Single Mode Fibers

Optical fibers used in communication systems can be broadly classified according to their ability to maintain polarization during transmission. The fiber most commonly deployed is the single-mode fiber (SMF), shown in [fig. 3.4](#). The SMF is an optical fiber designed to support the propagation of a single spatial mode of light. This is achieved by using a very small core diameter (8–10 μm), which is surrounded by a cladding with a slightly lower refractive index. The fiber used in this work also has an outer protective jacket with a diameter of $\varnothing 900, \mu\text{m}$. This value refers to the external jacket diameter and should not be confused with the much smaller optical core diameter. The refractive index difference is what confines the light within the fiber through total internal reflection. Due to its support for only a single spatial mode, SMFs eliminate modal dispersion, which is a phenomenon in which different spatial modes arrive at different times. Therefore, SMFs provide lower distortion, higher bandwidth, and longer transmission distances than multi-mode fibers. This also makes them the ideal transmission medium for modern optical communication systems and quantum networks. Moreover, in quantum communication systems, photons must remain indistinguishable

to achieve high interference visibility and reliable Bell-state measurements. That being said, the presence of multiple spatial modes can introduce timing uncertainty and mode mismatch, which in turn degrades quantum interference.



Figure 3.4: Commercial single-mode optical fiber manufactured by Thorlabs.

However, one of the main drawbacks of single mode fibers is that they generally do not preserve the polarization state of light. In an ideal perfectly symmetric fiber, orthogonal polarization components would propagate identically. However, in practice, SMFs exhibit small geometric imperfections, manufacturing asymmetries, and environmental disturbances. These effects will then introduce birefringence, causing different polarization components to drift from their original states.

3.3.3 Polarization-Maintaining Single Mode Fiber

As we saw previously, a normal single-mode fiber supports one spatial mode but that one spatial mode still has two polarization components that can mix because of the random birefringence caused by bending, stress, temperature fluctuations, and imperfections. However, polarization-maintaining fiber is a special SMF in which the fiber is intentionally made birefringent. For instance, Thorlabs mentions PANDA and Bow-Tie

polarization-maintaining SMFs.



Figure 3.5: Commercial polarization maintaining single-mode optical fiber manufactured by Thorlabs.

PM fibers have two built-in polarization axes: the fast axis and the slow axis. The fast axis has a lower effective refractive index, so light can travel faster through it. The slow axis has a higher effective refractive index, and thus light travels slower through it. In such cases, if linearly polarized light is sent exactly along one of the axes, most of the optical power stays on that same axis. This means that the output polarization remains stable instead of randomly rotating. However, it should be noted that PM fibers do not always preserve every possible input polarization state. This holds true only when the input polarization is properly aligned with one of the fiber axes. The addition of birefringence helps because, in SMFs, Δn of $\Delta n = n_x - n_y$ is small, random, and changes with the environment. PM fibers have much larger Δn . This strong birefringence separates the polarization modes so much that random coupling between them cancels out over distance. The phase difference accumulated between the two axes is given by:

$$\Delta\phi = \frac{2\pi}{\lambda} (n_{\text{slow}} - n_{\text{fast}}) L. \quad (3.12)$$

This phase difference grows as the light propagates. The distance over which the two axes accumulate a 2π phase difference is known as the beat length, given by the formula:

$$L_b = \frac{\lambda}{n_{\text{slow}} - n_{\text{fast}}}. \quad (3.13)$$

Therefore, a shorter beat length L_b means stronger birefringence, which also indicates better polarization-maintaining behavior.

3.4 Polarization Control Hardware

When we talk about polarization control hardware, we refer to the different types of waveplate that we use in the laboratory to change the polarization state of light. As mentioned in [section 2.3.2](#), light can be linearly, circularly, or elliptically polarized (although we will not deal with elliptical polarization within the scope of this thesis). Moreover, as we saw in [section 2.4](#) and [section 2.3.3](#), the polarization state depends on the relative amplitudes and relative phases of two orthogonal polarization components.

However, how do we implement this physically? The optical devices that allow us to modify the polarization state of incoming light are known as waveplates. The two most common waveplate configurations are the quarter-waveplate and the half-waveplate. These waveplates can be found in the form of manual devices or motorized devices that can be controlled using external systems such as computers. This section discusses these devices in detail.

3.4.1 Waveplates

Before we first see the mathematical description of polarization changes due to the rotation of waveplates, let us first draw a connection between actual waveplates and the Jones Vectors. In [section 2.4](#), we represent a Jones Vector by:

$$\mathbf{J} = \begin{bmatrix} A_x e^{i\phi_x} \\ A_y e^{i\phi_y} \end{bmatrix}.$$

Thus, any optical element that changes the polarization state can be represented by the Jones matrix M , where the output polarization state is given by:

$$J_{\text{out}} = M J_{\text{in}}$$

For waveplates, the matrix M simply becomes a quarter waveplate or a half waveplate matrix. Let us take the half-waveplate (HWP) for instance, the Jones matrix of a HWP is:

$$J(\text{HWP}, \theta) = i \begin{bmatrix} \cos 2\theta & \sin 2\theta \\ \sin 2\theta & -\cos 2\theta \end{bmatrix}$$

where θ is the angle between the incoming polarization basis and the waveplate's fast axis. In other words, a half waveplate introduces a phase difference of π between the fast and slow axis. Hence, if we had vertically polarized light V passing through a HWP, then the Jones matrix representation will be:

$$J(\text{HWP}, \theta) = i \begin{bmatrix} \cos 2\theta & \sin 2\theta \\ \sin 2\theta & -\cos 2\theta \end{bmatrix} \quad (3.14)$$

$$|V\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad (3.15)$$

$$J(\text{HWP}, \theta) |V\rangle = i \begin{bmatrix} \sin 2\theta \\ -\cos 2\theta \end{bmatrix} \quad (3.16)$$

From here we can substitute the θ rotations of the fast and slow axis in [eq. \(3.16\)](#) and determine the polarization state of the output light. For instance, let us take a look at the output state when we have a vertically polarized light as our input state and it passes through 0° and 45°

$$\theta = 0^\circ$$

$$J(\text{HWP}, 0^\circ) |V\rangle = i \begin{bmatrix} \sin 0^\circ \\ -\cos 0^\circ \end{bmatrix} = i \begin{bmatrix} 0 \\ -1 \end{bmatrix} \equiv \begin{bmatrix} 0 \\ 1 \end{bmatrix} = |V\rangle$$

$$\theta = 45^\circ$$

$$J(\text{HWP}, 45^\circ)|V\rangle = i \begin{bmatrix} \sin 90^\circ \\ -\cos 90^\circ \end{bmatrix} = i \begin{bmatrix} 1 \\ 0 \end{bmatrix} \equiv \begin{bmatrix} 1 \\ 0 \end{bmatrix} = |H\rangle$$

Likewise, the Jones matrix representation of a QWP is given by:

$$J(\text{QWP}, \theta) = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 + i \cos 2\theta & i \sin 2\theta \\ i \sin 2\theta & 1 - i \cos 2\theta \end{bmatrix}.$$

Now, if we follow the same previous example and assume a vertically polarized input light, then our transformation matrix becomes:

$$J(\text{QWP}, \theta) |V\rangle = \frac{1}{\sqrt{2}} \begin{bmatrix} i \sin 2\theta \\ 1 - i \cos 2\theta \end{bmatrix}. \quad (3.17)$$

Next we can substitute the θ rotations of the fast and slow axes in [eq. \(3.17\)](#) and determine the polarization state of the output light. The following table summarizes the output polarization state of a vertically polarized light when it passes through HWP and QWP.

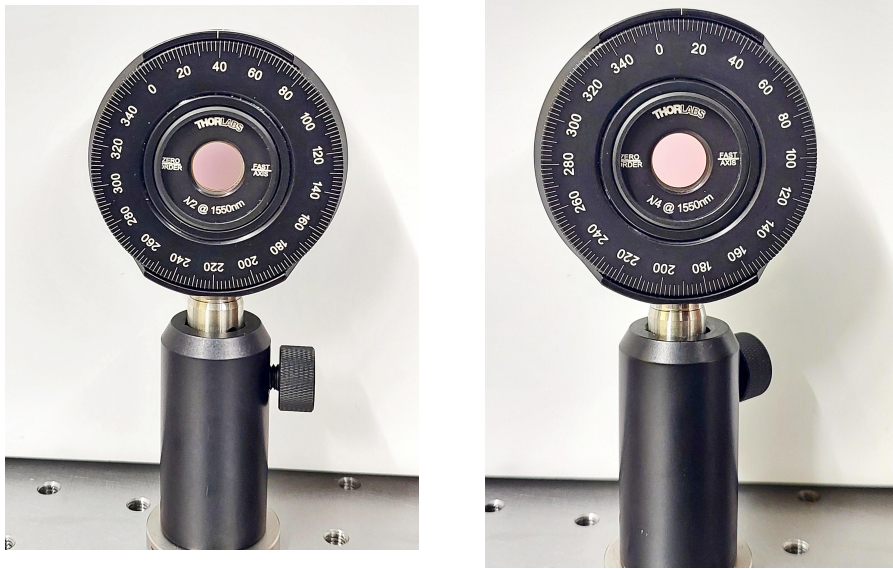


Figure 3.6: Left: half-waveplate. Right: quarter-waveplate.

Table 3.1: Polarization states for vertically polarized input light passing through HWP and QWP at different orientations.

θ	HWP Output	QWP Output
0°	V-polarization	V-polarization
22.5°	A-polarization	Elliptical R-polarization
45°	H-polarization	R-polarization
67.5°	D-polarization	Elliptical R-polarization
90°	V-polarization	V-polarization

3.4.2 Motorized Polarization Controller

The Thorlabs MPC320 Motorized Polarization Controller, shown in [fig. 3.7](#), is a three-paddle (two-paddle in the case of the MPC220) fiber polarization controller that manipulates the polarization state of light propagating through an optical fiber. Contrary to free-space polarization control, which uses QWPs and HWPs, the MPC320 induces birefringence directly within the fiber by wrapping the fiber around rotating paddles. The device has three independently motorized paddles and is designed for use with fibers having an outer jacket diameter of $\varnothing 900\mu\text{m}$, such as the single-mode fiber described in [section 3.3.2](#).

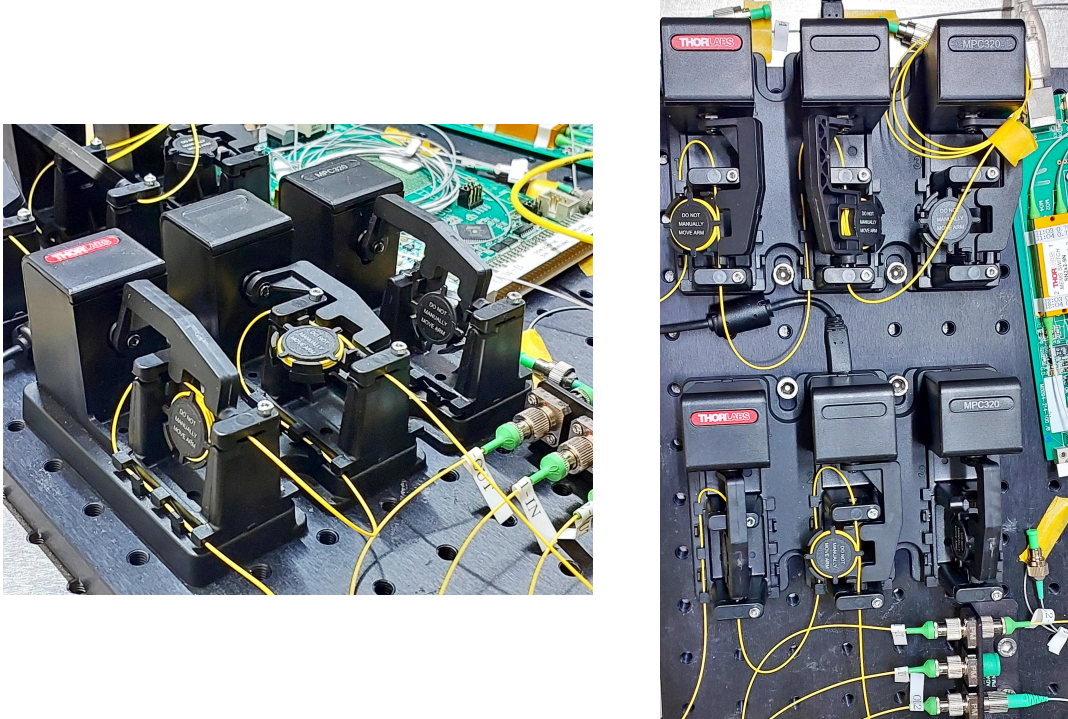


Figure 3.7: Thorlabs MPC320 motorized polarization controller

The MPC320 works by using stress-induced birefringence. When an optical fiber is wrapped around a circular spool, mechanical stress is introduced into the fiber. This stress creates a difference in the refractive index along the two orthogonal polarization axes, which produces birefringence. Birefringence depends on the diameter of the fiber cladding d , the diameter of the loop D , the number of fiber loops N , and the optical wavelength λ . The MPC320 can also act as a $QWP - HWP - QWP$ configuration, where the three-paddle controller can be treated as three waveplates connected in series. Thus, the first QWP converts an arbitrary input state into a linear polarization state, followed by the HWP which rotates that linear polarization. Finally, the second QWP converts the rotated linear state into the desired output polarization state.

However, during this work, we utilized only two of the three paddles. The optical fiber was routed only through the first two paddles. The third paddle was not loaded with fiber and therefore did not participate in the polarization compensation algorithm. Mathematically, this can be represented as:

$$|\psi_{\text{out}}\rangle = \mathbf{J}_{\text{QWP}}(\theta_2) \mathbf{J}_{\text{QWP}}(\theta_1) |\psi_{\text{in}}\rangle.$$

The exact utilization of MPC in this experiment, as well as the fiber installation process, will be discussed in the next chapter.

3.5 Beam Splitters

A beam splitter, shown in [fig. 3.8](#), is an optical device that divides an incoming light beam into two separate output paths. It is one of the fundamental components in optical communication systems, interferometers, and quantum optics experiments. It acts on the probability amplitude rather than the optical power. Therefore, a single photon that is incident on a 50:50 beam splitter does not split into two half-energy photons. Instead, the photon enters a quantum superposition of being transmitted and reflected. This can be represented as:

$$|1\rangle_a \rightarrow \frac{1}{\sqrt{2}} (|1\rangle_c + i |1\rangle_d)$$

where c and d denote the output ports and i corresponds to the phase shift associated with reflection. The most commonly used beam splitter in quantum optics is the 50:50

beam splitter (BS). The mode transformation can be written as:

$$\begin{pmatrix} \hat{c} \\ \hat{d} \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & i \\ i & 1 \end{pmatrix} \begin{pmatrix} \hat{a} \\ \hat{b} \end{pmatrix} \quad (3.18)$$

where $\hat{a}, \hat{b}$ are the input modes and $\hat{c}, \hat{d}$ are the output modes. Another type of beam splitter is the polarizing beam splitter (PBS). This type of beam splitter separates the incoming light according to its polarization state. This is done by transmitting one polarization component while reflecting the orthogonal polarization component:

$$|\psi\rangle = \alpha |H\rangle_T + \beta |V\rangle_R$$

In cases where we have a diagonally polarized photon:

$$|D\rangle = \frac{1}{\sqrt{2}} (|H\rangle + |V\rangle)$$

then after passing through the PBS, we will get a 50% probability of detecting the photon at the transmitted port and 50% probability of detecting at the reflected port:

$$|D\rangle \rightarrow \frac{1}{\sqrt{2}} (|H\rangle_T + |V\rangle_R).$$

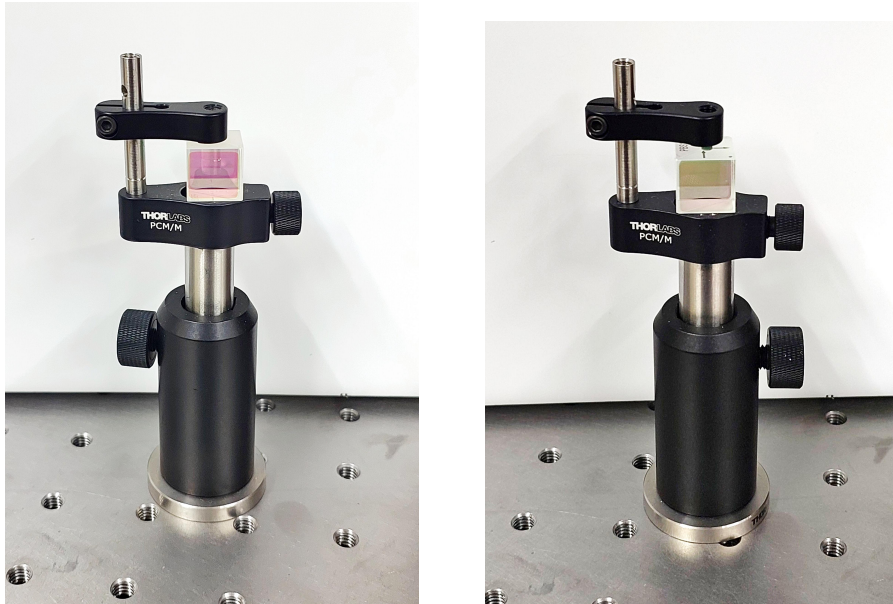


Figure 3.8: Left: beam splitter (BS). Right: polarizing beam splitter (PBS).

3.6 Fiber Coupler

A fiber coupler, shown in [fig. 3.9](#), is an optical device that is used to split or combine optical signals propagating within optical fibers. It has a similar role to that of beam splitters in free-space optics, however, the coupling process occurs within the fiber. Fiber couplers are characterized by their splitting ratio. For instance, a 50 : 50 coupler can be represented by:

$$P_1 = 0.5P_{in} \qquad P_2 = 0.5P_{in}$$

where P_{in} is the input power and P_1 and P_2 are the output powers such that $P_1 + P_2 = P_{in}$. There are also other splitting ratios such as 70 : 30, 90 : 10, 99 : 1, etc.

A 50 : 50 fiber coupler performs the same mathematical operation as the 50:50 beam splitter described in [eq. \(3.18\)](#). Therefore, it is a very useful optical device for HOM interference, Bell-state measurements, and entanglement swapping. Fiber couplers can also be classified on the basis of their input and output ports. For instance, there are 1×2 couplers that have one input and two output ports, and likewise 2×1 couplers that have two inputs and one output port. In addition, there are 2×2 couplers that are commonly used in interferometers.

However, an important fiber coupler relevant to this thesis is the fiber-based polarization beam splitter. This device is different from a normal coupler that splits according to a power ratio, as it instead separates light according to the polarization state of the incoming signal. Internally, the device contains a polarization beam splitter that is used to separate or combine orthogonal polarization states.

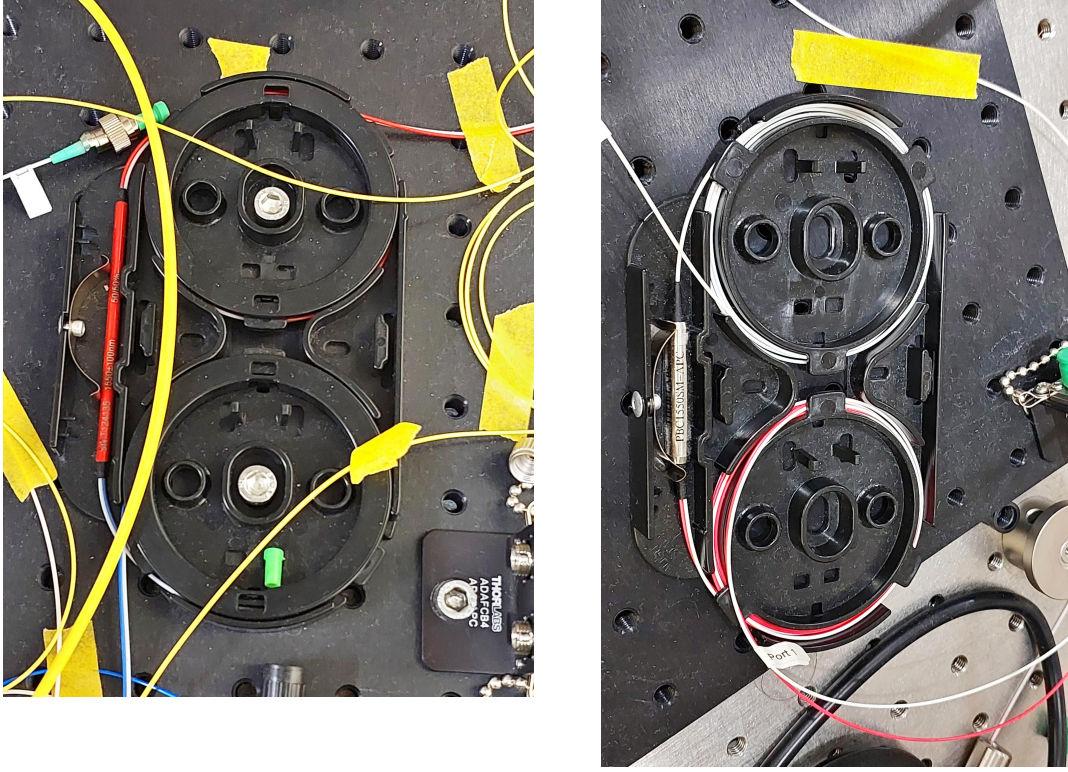


Figure 3.9: Left: fiber based beam splitter (FBS). Right: fiber based polarizing beam splitter (FPBS).

3.7 Superconducting Nanowire Single Photon Detector

The Superconducting Nanowire Single-Photon Detector (SNSPD) is an optical and infrared single-photon detector that operates using a superconducting nanowire biased close to its critical current. Mostly it consists of a superconducting nanowire, a meandering pattern, a bias current source, and a cryogenic cooling system. It is fabricated from materials such as Niobium Nitride (NbN), Tungsten Silicide (WSi), Molybdenum Silicide (MoSi), and so on. [Figure 3.10](#) shows the cryostat containing the detector, together with the controller and the associated amplification and time-tagging electronics.



Figure 3.10: Experimental SNSPD system, including the cryostat, controller, amplifier, and time-tagging electronics.

The operating principle starts with the nanowire being cooled below its critical temperature so that it becomes superconducting. Then, a bias current is applied close to the critical current. In the next step, a photon is absorbed, which transfers energy to the superconducting nanowire. This results in the formation of a hotspot. In the meantime, the absorbed energy locally destroys the superconductivity and hence creates a small resistive region. The Current is then redistributed around the hotspot, which in turn increases the current density near the hotspot. As a result, the superconducting state collapses locally across the width of the nanowire and generates a voltage pulse. This pulse is then interpreted as the detection of a photon. After this cycle, the hotspot cools and the wire returns to its original superconducting state.

This chapter introduced the experimental platform and main optical devices used in this work, including the entangled photon-pair source, optical fibers, motorized polarization controller, beam-splitting components, and single-photon detectors. With these components established, the next chapter describes how they are connected in the experimental setup and explains the automated polarization-compensation procedure used to control the motorized polarization controller based on the measured photon-count rate.

Chapter 4

Experimental Setup and Polarization Compensation

This chapter describes the complete experimental platform developed for adaptive polarization compensation. It explains the optical signal path, the motorized polarization controller configuration, the photon-detection system, the software-control architecture, and the optimization procedure used in the experiment.

4.1 Experimental Setup Overview

The experimental platform integrates photon-pair generation, polarization control, photon detection, and software-based feedback into a single system. The optical signal path begins with an entangled photon-pair source (EPPS) based on periodically poled lithium niobate (PPLN). Through the nonlinear optical process described [section 3.2.1](#), the source generates correlated photon pairs that are coupled to an optical fiber and transmitted to the polarization-control stage.

In the current experimental setup, the generated photons are first directed through a motorized polarization controller (MPC). The MPC modifies the state of polarization of the transmitted photons by introducing controllable stress-induced birefringence within the optical fiber. By rotating its motorized paddles, the MPC changes the relative phase and orientation of the polarization components without requiring the optical path to be disconnected.

After passing through the MPC, the photons are directed to a fiber based polarizing

beam splitter (FPBS). The FPBS separates the incoming optical field into two orthogonal polarization components. The corresponding output ports are then connected to the channels of the superconducting nanowire single-photon detector (SNSPD) system. The photon counts measured at these channels provide information about the polarization state arriving at the receiver.

The electronic and software components form the feedback and control section of the platform. The electrical signals produced by the SNSPD system are processed by the detector electronics and accessed by a Raspberry Pi. A Python-based control program running on the Raspberry Pi reads the measured photon counts and communicates with the MPC. Based on the detected count rates, the program adjusts the angular positions of the MPC paddles to find the polarization configuration that produces the desired detector response.

The complete signal and control flow can therefore be summarized as follows:

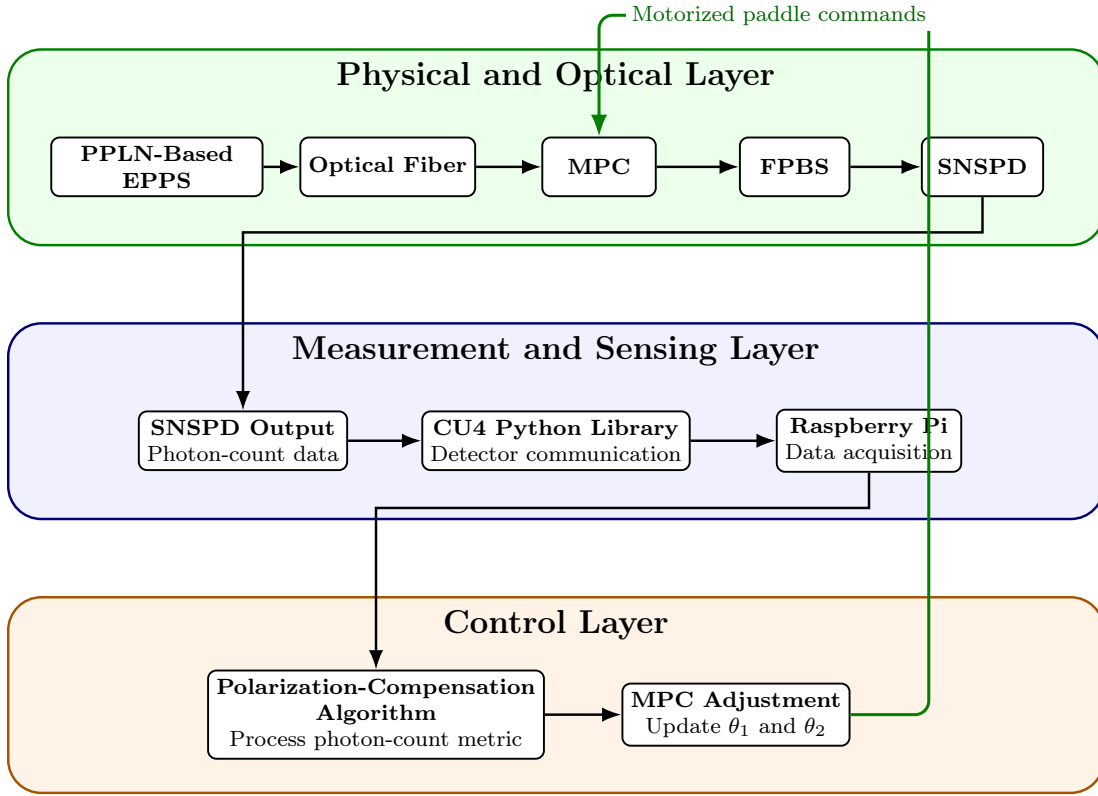


Figure 4.1: Three-layer architecture of the polarization-compensation experiment

This arrangement forms a closed-loop polarization compensation subsystem. The optical components are responsible for generating, manipulating, separating, and detecting photons, whereas the electronic and software components use the measured photon counts to control the polarization state.

4.2 Optical Signal Path

The optical signal path begins at the Type-II entangled photon-pair source. The internal construction and operation of the source are outside the scope of this chapter. However, for the present experiment, the source is treated as a node that produces pairs of photons with orthogonal polarization correlations.

The ideal two-photon polarization state may be written as:

$$|\Psi^+\rangle_{si} = \frac{1}{\sqrt{2}} (|H_s V_i\rangle + |V_s H_i\rangle), \quad (4.1)$$

where the subscripts s and i represent the signal and idler photons, respectively. After this, the photons are coupled into the optical fiber and transmitted toward the polarization-control system. An arbitrary polarization state traveling through one of the fiber paths can be written as:

$$|\psi_{\text{in}}\rangle = a |H\rangle + b |V\rangle, \quad |a|^2 + |b|^2 = 1. \quad (4.2)$$

In an ideal channel that preserves polarization, the coefficients a and b would remain unchanged apart from a common phase. However, in reality, bending, mechanical stress, temperature variation, and imperfections in the optical fiber introduce birefringence. The horizontal and vertical polarization components, therefore, accumulate different phases during propagation.

To describe this process, let us first rotate the $|\psi_{\text{in}}\rangle$ into the fiber's principal-axis basis by adding the term $\mathbf{R}[\alpha(t)]$, which is represented by:

$$\mathbf{R}(\alpha) = \begin{pmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{pmatrix}.$$

Thus, at this point, we will temporarily describe the photon using the fiber's fast and slow axis instead of horizontal and vertical polarization:

$$|\psi'\rangle = \mathbf{R}[\alpha(t)]|\psi_{\text{in}}\rangle.$$

Next, we will apply a birefringent phase delay in which the two orthogonal components

accumulate different phases:

$$\begin{aligned} E_{\text{fast}} &\rightarrow e^{i\delta/2} E_{\text{fast}}, \\ E_{\text{slow}} &\rightarrow e^{-i\delta/2} E_{\text{slow}}. \end{aligned}$$

Here, the important term is the relative phase difference between them which can be described by:

$$\frac{e^{i\delta/2}}{e^{-i\delta/2}} = e^{i\delta}.$$

The reason we use $+\delta/2$ and $-\delta/2$ here is because the average phase is removed and a common phase multiplying the entire quantum state does not change the measurable polarization properties. Therefore, the birefringent phase delay will be:

$$\begin{pmatrix} E'_{\text{fast}} \\ E'_{\text{slow}} \end{pmatrix} = \begin{pmatrix} e^{i\delta(t)/2} & 0 \\ 0 & e^{-i\delta(t)/2} \end{pmatrix} \begin{pmatrix} E_{\text{fast}} \\ E_{\text{slow}} \end{pmatrix}$$

and since the state in the fiber's principal-axis basis is given by $|\psi'\rangle = \mathbf{R}[\alpha(t)] |\psi_{\text{in}}\rangle$, the state after applying the birefringent phase delay becomes:

$$\begin{pmatrix} e^{i\delta(t)/2} & 0 \\ 0 & e^{-i\delta(t)/2} \end{pmatrix} \mathbf{R}[\alpha(t)] |\psi_{\text{in}}\rangle. \quad (4.3)$$

Finally, after applying the phase delay, we rotate the state back into the original horizontal-vertical coordinate system using $\mathbf{R}[-\alpha(t)]$. Therefore, the polarization transformation introduced by the fiber can be represented by a Jones matrix:

$$\mathbf{J}_f(t) = \mathbf{R}[-\alpha(t)] \begin{pmatrix} e^{i\delta(t)/2} & 0 \\ 0 & e^{-i\delta(t)/2} \end{pmatrix} \mathbf{R}[\alpha(t)], \quad (4.4)$$

where $\alpha(t)$ represents the orientation of the birefringent axis and $\delta(t)$ represents the relative phase delay between them. The phase delay may be expressed as:

$$\delta(t) = \frac{2\pi}{\lambda} \Delta n(t) L, \quad (4.5)$$

where $\Delta n(t)$ is the difference between the effective refractive indices, L is the fiber length, and λ is the wavelength of the transmitted photon. This equation comes from the phase accumulated by light while it propagates through a material. For a light traveling a distance L through a material with refractive index n , the phase is:

$$\phi = \frac{2\pi}{\lambda} nL.$$

Moreover, a birefringent fiber has two orthogonal polarization axes with different refractive indices $n_1 \neq n_2$. Therefore, the two polarization components accumulate different phases:

$$\begin{aligned}\phi_1 &= \frac{2\pi}{\lambda} n_1 L \\ \phi_2 &= \frac{2\pi}{\lambda} n_2 L\end{aligned}$$

and since $\delta = \phi_1 - \phi_2$, we get:

$$\begin{aligned}\delta &= \frac{2\pi}{\lambda} n_1 L - \frac{2\pi}{\lambda} n_2 L \\ \delta(t) &= \frac{2\pi}{\lambda} (n_1 - n_2) L.\end{aligned}\tag{4.6}$$

Defining the birefringence as Δn , we obtain the phase delay described in [eq. \(4.5\)](#). The exact polarization state arriving at the motorized polarization controller is therefore:

$$|\psi_f(t)\rangle = \mathbf{J}_f(t) |\psi_{in}\rangle.\tag{4.7}$$

For a lossless and fixed fiber transformation, the output remains a pure state. The fiber changes the polarization orientation, but does not necessarily destroy the state. However, the fiber transformation may vary during the measurement interval. Therefore, we need to express the observed state using a time averaged density matrix.

The input density matrix is given by ρ_{in} and will represent the polarization state entering the fiber. For a pure state $|\psi_{in}\rangle$, the density matrix is $|\psi_{in}\rangle \langle \psi_{in}|$. For instance, for a vertically polarized photon $|V\rangle$, the density matrix is:

$$\rho_V = |V\rangle \langle V| = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \begin{pmatrix} 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}.$$

Thus, the corresponding transformation of our fiber transformation discussed [eq. \(4.7\)](#) is:

$$\rho_f(t) = \mathbf{J}_f(t) \rho_{\text{in}} \mathbf{J}_f^\dagger(t).$$

$\mathbf{J}_f^\dagger(t)$ means the conjugate transpose of $\mathbf{J}_f(t)$. Therefore, the experimentally observed state can then be represented by the time-averaged density matrix:

$$\bar{\rho}_f = \frac{1}{T} \int_0^T \mathbf{J}_f(t) \rho_{\text{in}} \mathbf{J}_f^\dagger(t) dt, \quad (4.8)$$

where T is the detector acquisition interval. If several different polarization transformations occur during this interval, the measured polarization state may appear mixed such that:

$$\text{Tr}(\bar{\rho}_f^2) < 1. \quad (4.9)$$

The motorized polarization controller applies an additional controllable transformation to the incoming state. In the current experiment, the transformation depends on the angular positions of the two active paddles and can be written as:

$$\mathbf{J}_{\text{MPC}} = \mathbf{J}_{\text{MPC}}(\theta_1, \theta_2), \quad (4.10)$$

where θ_1 and θ_2 are the angles of the paddle. The polarization state after the MPC is therefore represented as:

$$\rho_{\text{out}} = \mathbf{J}_{\text{MPC}}(\theta_1, \theta_2) \bar{\rho}_f \mathbf{J}_{\text{MPC}}^\dagger(\theta_1, \theta_2). \quad (4.11)$$

The objective of the compensation system is to determine the paddle-angles that will make up for the fiber-induced polarization transformation. Ideally, the combined transformation satisfies:

$$\mathbf{J}_{\text{MPC}}(\theta_1, \theta_2) \mathbf{J}_f(t) \approx \mathbf{J}_{\text{target}}, \quad (4.12)$$

where $\mathbf{J}_{\text{target}}$ represents the desired polarization transformation at the receiver. After leaving the MPC, the photons enter a fiber based polarizing beam splitter (FPBS) discussed in [section 3.6](#). The FPBS projects the incoming state onto two orthogonal polarization components. For horizontal and vertical polarization measurements, the corresponding projection operators are:

$$\mathbf{\Pi}_H = |H\rangle\langle H|, \quad \mathbf{\Pi}_V = |V\rangle\langle V|. \quad (4.13)$$

The probabilities of the photon leaving through the two FPBS output ports are:

$$P_H = \text{Tr}(\mathbf{\Pi}_H \rho_{\text{out}}), \\ P_V = \text{Tr}(\mathbf{\Pi}_V \rho_{\text{out}}).$$

Tr is just the trace of our density matrix and, for a properly normalized density matrix, $\text{Tr}(\rho) = 1$. For instance, for a vertically polarized light:

$$|V\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

the density matrix is given by:

$$\rho_V = |V\rangle\langle V| = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \begin{pmatrix} 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$$

and the trace is the sum of the diagonal elements, $\text{Tr}(\rho_v) = 0 + 1 = 1$.

Finally, the FPBS outputs are connected to the SNSPD detector. The detected photon count on channel k during the acquisition time period can be approximated as:

$$C_k \approx \eta_k \Phi T P_k + B_k, \quad (4.14)$$

where η_k is the total transmission and detection efficiency of the channel k , Φ is the incident photon flux, T is the acquisition time, P_k is the corresponding polarization-projection probability and B_k represents background counts.

When the MPC is not adjusted, the polarization arriving at the FPBS may be poorly aligned with the selected detector channel. As a result, only a small fraction of the transmitted photons may be directed toward that SNSPD channel. Adjusting the MPC paddle angles changes the polarization projection probabilities and can therefore increase the detected count in the target channel.

The MPC does not generate additional photons. It just transforms the polarization state so that a larger fraction of the photons is directed toward the desired FPBS output. Therefore, optical losses that occur along the path must be distinguished from polarization drift. The distinction is that loss reduces the total number of transmitted photons, whereas polarization misalignment changes how the remaining photons are distributed between the FPBS outputs.

4.3 Polarization Compensation Configuration

This section describes how polarization compensation was implemented within the experimental setup. As mentioned in [section 4.1](#), The motorized polarization controller was installed in the optical path before the FPBS and the SNSPD. Thus, the polarization state modified by the optical fiber first enters the MPC, after which the compensated state is projected onto the horizontal and vertical polarization outputs of the FPBS.

The MPC320 contains three independently motorized paddles. However, only two paddles were used in the current experiment. The optical fiber was looped around the first two paddles, while the third paddle was not loaded with fiber, and therefore did not participate in the polarization transformation. The experimental MPC configuration can therefore be treated as a two-paddle polarization controller.

Each active paddle acts as a controllable fiber retarder. When the fiber is wrapped around a paddle spool, the bending stress introduces birefringence into the fiber. This produces two effective orthogonal axes with different refractive indices and therefore introduces a relative phase delay between the corresponding polarization components.

It is important to distinguish between the effect of looping the fiber and the effect

of rotating the paddle. The fiber loop around the spool produces the stress-induced birefringence and determines the approximate retardance of the paddle. The motorized rotation changes the orientation of the effective fast and slow axes relative to the incoming polarization state. Therefore, the angle of the paddle does not determine whether the paddle acts as a quarter-wave or half-wave retarder. Instead, effective retardance depends on wavelength, fiber cladding diameter, spool diameter, and number of fiber loops.

The transformation produced by an individual paddle can be represented as:

$$\mathbf{J}_k(\theta_k, \phi_k) = \mathbf{R}(-\theta_k) \begin{pmatrix} e^{i\phi_k/2} & 0 \\ 0 & e^{-i\phi_k/2} \end{pmatrix} \mathbf{R}(\theta_k), \quad (4.15)$$

where k represents the paddle, θ_k is its motorized rotation angle, and ϕ_k is the effective retardance introduced by the fiber loop. The rotation matrix is given by:

$$\mathbf{R}(\theta_k) = \begin{pmatrix} \cos \theta_k & -\sin \theta_k \\ \sin \theta_k & \cos \theta_k \end{pmatrix}. \quad (4.16)$$

The central diagonal matrix in [eq. \(4.15\)](#) introduces a relative phase difference ϕ_k between the two principal polarization components. The rotation matrices transform the polarization state into and out of the principal-axis basis of the paddle.

The retardance introduced by each fiber loop can be estimated as:

$$\phi_k = \frac{2\pi^2 a N_k d^2}{\lambda D}, \quad (4.17)$$

where $a = 0.133$ for silica fiber, N_k is the number of loops around the paddle k , d is the diameter of the fiber cladding, λ is the optical wavelength, and D is the diameter of the loop.

In the present experiment, a single-mode P3-SMF28Y-FC-1 patch cable with a $900\,\mu\text{m}$ outer jacket was used. The corresponding fiber cladding diameter was approximately $125\,\mu\text{m}$. Three complete loops were installed around each of the two active paddles, and the diameter of the MPC320's paddle loop was 18 mm. For an EPPS output wavelength of approximately 1560 nm, the retardance of each paddle is estimated as:

$$\phi_k = \frac{2\pi^2(0.133)(3)(125 \times 10^{-6})^2}{(1560 \times 10^{-9})(18 \times 10^{-3})} \approx 4.38 \text{ rad.} \quad (4.18)$$

This value is close to:

$$\frac{3\pi}{2} \approx 4.71 \text{ rad,} \quad (4.19)$$

which corresponds to a multi-order quarter-wave retardance. Therefore, each active paddle can be treated approximately as a quarter-wave retarder, and the experimental arrangement can be represented as a QWP–QWP configuration.

Since two paddles were active, the complete MPC transformation is given by:

$$\mathbf{J}_{\text{MPC}}(\theta_1, \theta_2) = \mathbf{J}_2(\theta_2, \phi_2)\mathbf{J}_1(\theta_1, \phi_1), \quad (4.20)$$

where

$$\phi_1 \approx \phi_2 \approx 4.38 \text{ rad.} \quad (4.21)$$

The order of the matrices follows the order in which the photons encounter the paddles. The transformation of the first paddle is applied first and therefore appears on the right-hand side of the matrix product.

Since both active paddles behave approximately as quarter-wave retarders, the MPC transformation may also be written as:

$$\mathbf{J}_{\text{MPC}}(\theta_1, \theta_2) \approx \mathbf{J}_{\text{QWP}}(\theta_2)\mathbf{J}_{\text{QWP}}(\theta_1). \quad (4.22)$$

The approximation symbol is used because the estimated retardance of each paddle is close, but not exactly the ideal quarter-wave value.

Because the third paddle was not loaded with fiber, it was not part of the optical path. If the system is formally represented as a three-paddle transformation, the contribution of the third paddle can be written as the identity matrix ($\mathbf{J}_3 = \mathbf{I}$) and therefore does not change the two-paddle transformation. For an instantaneous pure polarization state, the state leaving the MPC is:

$$|\psi_{\text{MPC}}\rangle = \mathbf{J}_{\text{MPC}}(\theta_1, \theta_2) |\psi_{\text{f}}\rangle, \quad (4.23)$$

where $|\psi_{\text{f}}\rangle$ is the polarization state arriving from the optical fiber. Combining the fiber and MPC transformations gives:

$$|\psi_{\text{MPC}}\rangle = \mathbf{J}_2(\theta_2, \phi_2) \mathbf{J}_1(\theta_1, \phi_1) \mathbf{J}_{\text{f}}(t) |\psi_{\text{in}}\rangle. \quad (4.24)$$

Using the density-matrix representation, the state after the MPC can be written as:

$$\rho_{\text{MPC}} = \mathbf{J}_{\text{MPC}} \rho_{\text{f}} \mathbf{J}_{\text{MPC}}^\dagger. \quad (4.25)$$

Therefore, The two paddle-angles, θ_1 and θ_2 , determine the polarization state presented to the FPBS. Rotating either paddle changes the orientation of its effective birefringent axis and consequently modifies the relative amplitudes and phases of the horizontal and vertical polarization components.

The two paddle transformations are coupled in their effect on the final polarization state. The output of the first paddle becomes the input of the second paddle, which means that a change in θ_1 can alter the value of θ_2 required to obtain the desired output. This coupling motivates the use of a two-dimensional optimization procedure instead of optimizing each paddle independently.

For compensation toward the vertical FPBS output, the desired paddle angles are those that maximize the vertical projection probability:

$$P_V(\theta_1, \theta_2) = \text{Tr} \left(\hat{\Pi}_V \rho_{\text{MPC}} \right), \quad (4.26)$$

where

$$\hat{\Pi}_V = |V\rangle \langle V| \quad (4.27)$$

is the vertical-polarization projection operator. For a pure polarization state, the same probability can be expressed as:

$$P_V(\theta_1, \theta_2) = \langle \psi_{\text{MPC}} | \hat{\Pi}_V | \psi_{\text{MPC}} \rangle. \quad (4.28)$$

By increasing this projection probability, we can direct a larger fraction of the transmitted photons toward the vertical output of the FPBS and therefore increase the count measured by the corresponding SNSPD channel. In this operation, the compensation algorithm varies θ_1 and θ_2 within the available paddle rotation range of 0° to 160° . The measured photon count is used as experimental feedback for the resulting polarization projection, and the two paddle angles are updated jointly until the selected detector response is optimized.

4.4 Software Control Architecture

The Raspberry Pi serves as the control computer for the experimental platform. A Python program running on the Raspberry Pi communicates with both the SNSPD detector electronics and the motorized polarization controller. These two communication paths are implemented using separate Python libraries: the CU4 control library is used to acquire photon-count measurements, while the PnPQ library is used to control the angular positions of the MPC paddles.

The overall software communication path can be summarized as:

$$\begin{aligned} \text{SNSPD} &\rightarrow \text{CU4 Control Unit} \rightarrow \text{CU4 Python Library} \\ &\rightarrow \text{Raspberry Pi} \rightarrow \text{PnPQ} \rightarrow \text{MPC320}. \end{aligned} \quad (4.29)$$

Although the two libraries communicate with different hardware interfaces, they are accessed by the same polarization-compensation program. This allows paddle movement, detector measurement, and paddle adjustment to be performed within a single software loop.

4.4.1 Communication with the SNSPD System

The SNSPD is connected to a Scontel CU4 control unit, which provides the electronic interface required to operate the detector and access its measured photon counts. In addition, communication between the Raspberry Pi and the CU4 control unit is established through the local Ethernet network using the CU4 Python library. Once the

CU4 unit is located, the program selects the required detector channel of the SNSPD.

The counter time constant defines the measurement interval where photon-detection events are accumulated. For each MPC configuration, the CU4 counter accumulates photon-detection events according to its configured time constant. The program then requests the corresponding photon count through the CU4 Python library. However, the detector occasionally returns an invalid or incomplete response. Hence, the control program checks the received detector value before using it. If a valid count is not obtained, the measurement request is repeated instead of passing an invalid value to the optimization procedure.

```
1 import CU4lib as cu
2
3 def connect_channel(channel, cu4_address):
4     unit = cu.CU4List()[cu4_address]
5     sdm = unit[channel]
6     sdm.init()
7     return sdm
8
9 sdm = connect_channel(channel, CU4_ADDRESS)
10 counts = sdm.counter.counts
```

Listing 4.1: Communication with the SNSPD system through the CU4 control unit.

4.4.2 Communication with the MPC320

The MPC320 is connected directly to the Raspberry Pi via USB. Communication with the controller is implemented using the Python PnPQ library. PnPQ provides an interface for controlling laboratory optical devices, including the Thorlabs MPC320.

The MPC is identified by its serial number. The PnPQ library uses this serial number to locate the corresponding USB serial port and establish an APT communication connection with the controller. Moreover, each motorized paddle is represented as a separate control channel. In the present experiment, only channels 1 and 2 are moved because the optical fiber is installed only on the first two paddles. The third channel is therefore not included in the compensation procedure.

The desired angular position is sent to the MPC using an absolute-movement command:

$$\theta_k^{\text{new}} \longrightarrow \text{move_absolute}(k, \theta_k^{\text{new}}), \quad (4.30)$$

where $k \in \{1, 2\}$ identifies the active paddle. PnPQ converts the requested position from degrees into the internal motor-step representation used by the MPC controller. The movement command is treated as a blocking operation, meaning that the program waits until the commanded paddle position is reached before continuing to the detector measurement. This is very important because acquiring counts while a paddle is still rotating would associate the detector measurement with an undefined intermediate position.

```

1 from pnpq.apr.connection import AprConnection
2 from pnpq.apr.protocol import ChanIdent
3 from pnpq.units import pnpq_ureg
4 from pnpq.devices.polarization_controller_thorlabs_mpc import (
5     PolarizationControllerThorlabsMPC320
6 )
7
8 MPC_SERIAL = "<MPC_SERIAL_NUMBER>"
9
10 with AprConnection(serial_number=MPC_SERIAL) as connection:
11     mpc = PolarizationControllerThorlabsMPC320(
12         connection=connection
13     )
14
15     mpc.move_absolute(
16         ChanIdent.CHANNEL_1,
17         theta1 * pnpq_ureg.deg
18     )
19
20     mpc.move_absolute(
21         ChanIdent.CHANNEL_2,
22         theta2 * pnpq_ureg.deg
23     )

```

Listing 4.2: Communication with the MPC320 through the PnPQ library.

4.5 Polarization Compensation Algorithm

The polarization-compensation algorithm determines the angular positions of the two active motorized polarization-controller paddles that produce a high photon-count rate at the selected detector channel. The detector output is used directly as feedback, and no analytical model of the complete optical link is required. The optical system is therefore treated as a black-box function that maps the two paddle angles to a measured photon-count rate.

Let θ_1 and θ_2 denote the angular positions of the first and second active paddles, respectively. The measured photon-count rate is written as:

$$C = C(\theta_1, \theta_2), \quad (4.31)$$

where $C(\theta_1, \theta_2)$ represents the photon-count rate obtained after applying the selected paddle-angle configuration. The objective of the compensation algorithm is to determine:

$$(\theta_1^*, \theta_2^*) = \arg \max_{\theta_1, \theta_2} C(\theta_1, \theta_2). \quad (4.32)$$

subject to the physical paddle-angle constraints:

$$0^\circ \leq \theta_1, \theta_2 \leq 160^\circ. \quad (4.33)$$

The algorithm uses a hierarchical multiresolution search to reduce the optimization time instead of an exhaustive evaluation of every possible paddle-angle pair. At each resolution, nine angle configurations are measured, a two-dimensional quadratic response surface is fitted to the measurements, and the estimated maximum of that surface is used to determine the center of the next search region. When the fitted model is not considered reliable, the strongest directly measured point is used instead.

4.5.1 Hierarchical Search Structure

The search consists of five stages with progressively decreasing angular step sizes:

$$h_k = \frac{80^\circ}{2^{k-1}}, \quad k = 1, 2, \dots, 5. \quad (4.34)$$

At the k -th stage, the search is centered at:

$$\left(\theta_{1,c}^{(k)}, \theta_{2,c}^{(k)} \right), \quad (4.35)$$

where $\theta_{1,c}^{(k)}$ and $\theta_{2,c}^{(k)}$ are the current center angles for the first and second paddles. Three angular values are generated for each paddle:

$$\Theta_1^{(k)} = \left\{ \theta_{1,c}^{(k)} - h_k, \theta_{1,c}^{(k)}, \theta_{1,c}^{(k)} + h_k \right\}. \quad (4.36)$$

and

$$\Theta_2^{(k)} = \left\{ \theta_{2,c}^{(k)} - h_k, \theta_{2,c}^{(k)}, \theta_{2,c}^{(k)} + h_k \right\}. \quad (4.37)$$

The Cartesian product of these two sets produces a 3×3 grid:

$$\mathcal{G}_k = \Theta_1^{(k)} \times \Theta_2^{(k)}. \quad (4.38)$$

Therefore, each stage contains, $N_{\text{stage}} = 3 \times 3 = 9$ paddle-angle configurations. The initial search center is selected as the midpoint of the permitted paddle range:

$$\theta_{1,c}^{(1)} = \theta_{2,c}^{(1)} = \frac{0^\circ + 160^\circ}{2} = 80^\circ. \quad (4.39)$$

Since the first-stage step size is $h_1 = 80^\circ$, the first-stage angle values are:

$$\Theta_1^{(1)} = \Theta_2^{(1)} = \{0^\circ, 80^\circ, 160^\circ\}. \quad (4.40)$$

The first stage therefore samples the entire permitted angle range using the nine configurations:

$$\mathcal{G}_1 = \left\{ \begin{array}{l} (0^\circ, 0^\circ), (0^\circ, 80^\circ), (0^\circ, 160^\circ), \\ (80^\circ, 0^\circ), (80^\circ, 80^\circ), (80^\circ, 160^\circ), \\ (160^\circ, 0^\circ), (160^\circ, 80^\circ), (160^\circ, 160^\circ) \end{array} \right\}. \quad (4.41)$$

Each subsequent stage uses half the previous angular step:

$$h_{k+1} = \frac{h_k}{2}. \quad (4.42)$$

Across the five stages, the algorithm evaluates 45 angle pairs. When final verification is enabled, an additional measurement is taken at the selected angle pair, bringing the total to 46 measurements.

Before each measurement, the required paddles are moved to the selected angular positions. A one-second settling period is applied whenever at least one paddle changes position. A single valid photon-count reading is then collected. Moreover, corrupted, non-finite, negative, or otherwise invalid detector responses are rejected and retried. In addition, the angle configurations are traversed using a snake-pattern scan. The second-paddle angles are visited in ascending order for one row and in descending order for the following row. This does not alter the mathematical grid, but reduces unnecessary motor movement between consecutive measurements.

4.5.2 Boundary Handling

A symmetric three-point search axis must remain within the permitted angular range. For a stage with step size h_k , the search center must therefore satisfy:

$$\theta_{\min} + h_k \leq \theta_c^{(k)} \leq \theta_{\max} - h_k, \quad (4.43)$$

where

$$\theta_{\min} = 0^\circ, \quad \theta_{\max} = 160^\circ. \quad (4.44)$$

If a requested center lies too close to either boundary, it is shifted according to:

$$\theta_{c,\text{actual}}^{(k)} = \text{clip}\left(\theta_{c,\text{requested}}^{(k)}, \theta_{\min} + h_k, \theta_{\max} - h_k\right). \quad (4.45)$$

For example, if the requested center is 160° and the next step size is 40° , the actual center becomes 120° . The resulting search points are then $80^\circ, 120^\circ, 160^\circ$. Shifting the center instead of clipping the three individual grid points preserves equal spacing and prevents repeated measurements.

```

1 ANGLE_MIN = 0.0
2 ANGLE_MAX = 160.0
3
4 def prepare_center_for_step(requested_center, step_size):
5     minimum_center = ANGLE_MIN + step_size
6     maximum_center = ANGLE_MAX - step_size
7
8     return float(
9         np.clip(
10             requested_center,
11             minimum_center,
12             maximum_center,
13         )
14     )

```

Listing 4.3: Implementation of boundary handling for the hierarchical search.

4.5.3 Normalized Search Coordinates

At each refinement stage, the photon-count rate is measured at nine paddle-angle combinations arranged on a 3×3 grid. The simplest approach would be to select the grid point with the highest measured response as the center of the next stage. However, this would restrict the estimated optimum to one of the nine discrete measurement positions. The true maximum may lie between these positions, while measurement noise may also cause a neighboring point to appear larger than the actual optimum.

Thus, to obtain a continuous estimate of the local maximum from the same nine measurements, the measured response is approximated using a quadratic response-surface model. The fitted model describes the local variation of the photon-count rate and allows the next search center to be placed between the measured grid points. This improves the angular resolution of the search without requiring additional measurements.

The quadratic response-surface model is expressed using normalized coordinates. At stage k , the normalized variables are defined as:

$$x = \frac{\theta_1 - \theta_{1,c}^{(k)}}{h_k}. \quad (4.46)$$

and

$$y = \frac{\theta_2 - \theta_{2,c}^{(k)}}{h_k}. \quad (4.47)$$

For the three paddle positions along each axis

$$\theta_c^{(k)} - h_k, \quad \theta_c^{(k)}, \quad \theta_c^{(k)} + h_k, \quad (4.48)$$

the corresponding normalized coordinates are:

$$-1, \quad 0, \quad 1. \quad (4.49)$$

Consequently, every search stage uses the same normalized 3×3 grid:

$$(x, y) \in -1, 0, 1 \times -1, 0, 1. \quad (4.50)$$

Normalization improves the numerical conditioning of the model-fitting procedure because the independent variables remain between -1 and 1 , regardless of the size of the physical step or the search center.

4.5.4 Quadratic Response-Surface Model

The nine measurements collected during one of the five stages are approximated using the two-dimensional quadratic model:

$$\hat{C}(x, y) = ax^2 + by^2 + cxy + dx + ey + f. \quad (4.51)$$

where $\hat{C}(x, y)$ is the photon-count rate predicted by the fitted model and

$$a, b, c, d, e, f \tag{4.52}$$

are the fitted coefficients.

The constant coefficient f represents the predicted response at the center of the search region because

$$\widehat{C}(0,0) = f. \tag{4.53}$$

The coefficients d and e describe the local slopes in the two normalized paddle-angle directions:

$$\left. \frac{\partial \widehat{C}}{\partial x} \right|_{(0,0)} = d. \tag{4.54}$$

and

$$\left. \frac{\partial \widehat{C}}{\partial y} \right|_{(0,0)} = e. \tag{4.55}$$

The coefficients a and b describe the curvature along the two directions of the paddle-angle:

$$\frac{\partial^2 \widehat{C}}{\partial x^2} = 2a. \tag{4.56}$$

and

$$\frac{\partial^2 \widehat{C}}{\partial y^2} = 2b. \tag{4.57}$$

The mixed coefficient c describes the interaction between the two paddles:

$$\frac{\partial^2 \widehat{C}}{\partial x \partial y} = c. \tag{4.58}$$

A non-zero value of c indicates that the effect of changing one paddle depends on the position of the other paddle. Geometrically, this term allows the principal axes of the fitted response surface to rotate relative to the θ_1 and θ_2 axes.

The quadratic model is not assumed to represent the complete polarization landscape over the entire angular range. Instead, it provides a local approximation of the measured response within the current 3×3 search region. This approximation is motivated by the second-order Taylor expansion of a smooth two-variable function around the current search center.

4.5.5 Least-Squares Estimation

For the i -th measured point in one search stage, the fitted model satisfies:

$$C_i \approx ax_i^2 + by_i^2 + cx_iy_i + dx_i + ey_i + f, \quad (4.59)$$

where C_i is the measured photon-count rate and (x_i, y_i) is the corresponding normalized angle pair. The nine measurement equations are written in matrix form as:

$$\mathbf{A}\boldsymbol{\beta} \approx \mathbf{C}, \quad (4.60)$$

where

$$\boldsymbol{\beta} = \begin{pmatrix} a \\ b \\ c \\ d \\ e \\ f \end{pmatrix} \quad (4.61)$$

is the unknown coefficient vector,

$$\mathbf{C} = \begin{pmatrix} C_1 \\ C_2 \\ \vdots \\ C_9 \end{pmatrix} \quad (4.62)$$

is the measurement vector, and the i -th row of the design matrix is

$$\mathbf{A}_i = \begin{pmatrix} x_i^2 & y_i^2 & x_i y_i & x_i & y_i & 1 \end{pmatrix}. \quad (4.63)$$

For the normalized grid $(x, y) \in \{-1, 0, 1\}^2$, one possible ordering of the complete design matrix is

$$\mathbf{A} = \begin{pmatrix} 1 & 1 & 1 & -1 & -1 & 1 \\ 1 & 0 & 0 & -1 & 0 & 1 \\ 1 & 1 & -1 & -1 & 1 & 1 \\ 0 & 1 & 0 & 0 & -1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 & 1 \\ 1 & 1 & -1 & 1 & -1 & 1 \\ 1 & 0 & 0 & 1 & 0 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 \end{pmatrix}. \quad (4.64)$$

Since the quadratic model contains six unknown coefficients, but nine measurements are available, the system is overdetermined. The coefficient vector is selected to minimize the sum of the squared fitting errors:

$$\hat{\boldsymbol{\beta}} = \arg \min_{\boldsymbol{\beta}} \|\mathbf{A}\boldsymbol{\beta} - \mathbf{C}\|_2^2. \quad (4.65)$$

which can also be written as:

$$\hat{\boldsymbol{\beta}} = \arg \min_{\boldsymbol{\beta}} \sum_{i=1}^9 \left[C_i - \hat{C}(x_i, y_i) \right]^2. \quad (4.66)$$

For a full-rank design matrix, the formal least-squares solution can be written as:

$$\hat{\boldsymbol{\beta}} = (\mathbf{A}^T \mathbf{A})^{-1} \mathbf{A}^T \mathbf{C}. \quad (4.67)$$

In the software implementation, as shown in [listing 4.4](#), the coefficient vector is obtained using a numerical least-squares solver rather than explicitly calculating the inverse matrix. This is done to improve numerical stability.

```

1 coefficients, residuals, rank, singular_values = np.linalg.lstsq(
2     design_matrix,
3     counts,
4     rcond=None,
5 )

```

Listing 4.4: Least-squares estimation of the quadratic response-surface coefficients from the measured photon-count rates using NumPy’s numerical solver.

The design matrix must also contain six linearly independent columns. The model is therefore rejected when $\text{rank}(\mathbf{A}) < 6$. Using nine measurements instead of the minimum of six provides additional constraints and reduces the sensitivity of the fitted coefficients to individual measurement fluctuations.

4.5.6 Model Fit Quality

After determining the quadratic coefficients, the fitted photon-count values are calculated as:

$$\hat{C} = \mathbf{A} \hat{\boldsymbol{\beta}}. \quad (4.68)$$

The residual associated with the i -th measurement is:

$$\varepsilon_i = C_i - \hat{C}(x_i, y_i). \quad (4.69)$$

The residual sum of squares is:

$$SS_{\text{res}} = \sum_{i=1}^9 \left[C_i - \hat{C}(x_i, y_i) \right]^2. \quad (4.70)$$

The total sum of squares is:

$$SS_{\text{tot}} = \sum_{i=1}^9 (C_i - \overline{C})^2, \quad (4.71)$$

where

$$\overline{C} = \frac{1}{9} \sum_{i=1}^9 C_i \quad (4.72)$$

is the mean count rate of the nine measurements. The coefficient of determination is then calculated as:

$$R^2 = 1 - \frac{SS_{\text{res}}}{SS_{\text{tot}}}. \quad (4.73)$$

A value of R^2 close to one indicates that the quadratic model closely follows the measured values. A low value indicates that the current response region is not well described by a quadratic surface or that the measurements contain substantial fluctuations. Therefore, The fitted model is accepted only when $R^2 \geq R_{\text{min}}^2$. The current polarization compensation algorithm uses $R_{\text{min}}^2 = 0.60$ as a threshold. This threshold is an algorithmic reliability parameter rather than a fundamental physical constant.

4.5.7 Prediction of the Quadratic Maximum

The gradient of the fitted response surface is:

$$\nabla \hat{C}(x, y) = \begin{pmatrix} \frac{\partial \hat{C}}{\partial x} \\ \frac{\partial \hat{C}}{\partial y} \end{pmatrix} = \begin{pmatrix} 2ax + cy + d \\ cx + 2by + e \end{pmatrix}. \quad (4.74)$$

A stationary point occurs where

$$\nabla \hat{C}(x^*, y^*) = \begin{pmatrix} 0 \\ 0 \end{pmatrix}. \quad (4.75)$$

The stationary-point equations are therefore:

$$2ax^* + cy^* + d = 0, \quad (4.76)$$

and

$$cx^* + 2by^* + e = 0. \quad (4.77)$$

These equations can be written as:

$$\mathbf{H} \begin{pmatrix} x^* \\ y^* \end{pmatrix} = -\mathbf{g}, \quad (4.78)$$

where

$$\mathbf{H} = \begin{pmatrix} 2a & c \\ c & 2b \end{pmatrix} \quad (4.79)$$

is the Hessian matrix, and

$$\mathbf{g} = \begin{pmatrix} d \\ e \end{pmatrix} \quad (4.80)$$

is the gradient evaluated at the normalized center $(0, 0)$. Provided that $\mathbf{H}$ is non-singular, the predicted stationary point is:

$$\begin{pmatrix} x^* \\ y^* \end{pmatrix} = -\mathbf{H}^{-1}\mathbf{g}. \quad (4.81)$$

The implementation solves the linear system in [eq. \(4.78\)](#) directly instead of explicitly calculating $\mathbf{H}^{-1}$.

The normalized prediction is converted back into physical paddle angles using:

$$\theta_{1,\text{pred}}^{(k)} = \theta_{1,c}^{(k)} + h_k x^*, \quad (4.82)$$

$$\theta_{2,\text{pred}}^{(k)} = \theta_{2,c}^{(k)} + h_k y^*. \quad (4.83)$$

The predicted count rate at this location is:

$$\widehat{C}_{\text{pred}}^{(k)} = \widehat{C}(x^*, y^*). \quad (4.84)$$

This value is a prediction of the model and is not treated as a physical measurement until the corresponding angle-pair is applied to the polarization controller and evaluated by the detector.

4.5.8 Hessian Curvature Test

The stationary point obtained from [eq. \(4.81\)](#) is not necessarily a maximum. It may also represent a minimum or a saddle point. Therefore, the curvature of the fitted surface is evaluated using the eigenvalues of the Hessian matrix. For instance, let the two eigenvalues of $\mathbf{H}$ be λ_1 and λ_2 . The fitted stationary point represents a local maximum only when the Hessian is negative definite:

$$\lambda_1 < 0 \quad \text{and} \quad \lambda_2 < 0. \quad (4.85)$$

For the two-dimensional Hessian in [eq. \(4.79\)](#), the equivalent conditions are $a < 0$ and $4ab - c^2 > 0$. These conditions indicate that the fitted response surface curves downward in every direction around the stationary point.

4.5.9 Hessian Conditioning

A Hessian matrix may be mathematically invertible but still be numerically unstable. This occurs when the fitted response surface is nearly flat in one direction. In that case, small photon-count fluctuations may cause a large change in the predicted stationary point. The numerical sensitivity of the Hessian is evaluated using its condition number:

$$\kappa(\mathbf{H}) = \|\mathbf{H}\|_2 \|\mathbf{H}^{-1}\|_2. \quad (4.86)$$

For a symmetric Hessian, the condition number is related to the ratio between the magnitudes of its largest and smallest eigenvalues. A large value indicates that the position predicted by the model may be highly sensitive to measurement noise. The model is rejected when $\kappa(\mathbf{H}) > \kappa_{\max}$. The current implementation uses $\kappa_{\max} = 10^8$.

4.5.10 Trust-Region Validation

As described in [section 4.5.3](#), the quadratic model is constructed only from measurements inside the normalized region:

$$-1 \leq x, y \leq 1. \quad (4.87)$$

The predicted stationary point is therefore accepted only when it remains inside this measured region:

$$|x^*| \leq 1 \quad \text{and} \quad |y^*| \leq 1. \quad (4.88)$$

This condition defines a trust region. Predictions inside the trust region are interpolations between or near measured points. Predictions outside the trust region would require extrapolation beyond the available measurements and are therefore rejected.

The corresponding physical angles must also satisfy:

$$0^\circ \leq \theta_{1,\text{pred}}^{(k)}, \theta_{2,\text{pred}}^{(k)} \leq 160^\circ. \quad (4.89)$$

4.5.11 Model Acceptance and Fallback Selection

After calculating all terms, the quadratic prediction is accepted only when the design matrix has a full column rank such that $\text{rank}(\mathbf{A}) = 6$, and the fitted model satisfies $R^2 \geq 0.60$. In addition, both eigenvalues of the Hessian must be negative, $\lambda_1 < 0$ and $\lambda_2 < 0$, which confirms that the predicted stationary point is a local maximum. The Hessian must also satisfy the conditioning requirement $\kappa(\mathbf{H}) \leq 10^8$. Finally, the predicted stationary point must also lie inside the current normalized search region, such that $|x^*| \leq 1$ and $|y^*| \leq 1$. After conversion to physical paddle angles, both predicted positions must remain within the permitted angular range:

$$0^\circ \leq \theta_{1,\text{pred}}^{(k)}, \theta_{2,\text{pred}}^{(k)} \leq 160^\circ. \quad (4.90)$$

The fitted coefficients and predicted quantities must also be finite numerical values. If all validation conditions are satisfied, the selected point for the next stage is:

$$\left(\theta_{1,\text{selected}}^{(k)}, \theta_{2,\text{selected}}^{(k)}\right) = \left(\theta_{1,\text{pred}}^{(k)}, \theta_{2,\text{pred}}^{(k)}\right). \quad (4.91)$$

If one or more conditions fail, the model prediction is discarded. The algorithm then selects the strongest directly measured configuration from the current stage:

$$\left(\theta_{1,\text{best}}^{(k)}, \theta_{2,\text{best}}^{(k)}\right) = \arg \max_{(\theta_1, \theta_2) \in \mathcal{G}_k} C(\theta_1, \theta_2). \quad (4.92)$$

The selected point becomes:

$$\left(\theta_{1,\text{selected}}^{(k)}, \theta_{2,\text{selected}}^{(k)}\right) = \left(\theta_{1,\text{best}}^{(k)}, \theta_{2,\text{best}}^{(k)}\right). \quad (4.93)$$

The fallback mechanism prevents an unreliable fitted model from moving the search into an unsupported region. Hence, every stage uses either a validated quadratic prediction or the strongest physical measurement obtained during that stage.

4.5.12 Selection of the Next Search center

For stages $k = 1, \dots, 4$, the selected point determines the requested center of the next stage:

$$\theta_{1,\text{c,requested}}^{(k+1)} = \theta_{1,\text{selected}}^{(k)}, \quad (4.94)$$

and

$$\theta_{2,\text{c,requested}}^{(k+1)} = \theta_{2,\text{selected}}^{(k)}. \quad (4.95)$$

The boundary-handling operation in [eq. \(4.45\)](#) is then applied using the next step size h_{k+1} . The resulting center is used to generate the following 3×3 search grid.

The search therefore follows the sequence:

$$\begin{aligned}
80^\circ : \quad & 3 \times 3 \text{ scan} \longrightarrow \text{quadratic fit and validation,} \\
40^\circ : \quad & 3 \times 3 \text{ scan} \longrightarrow \text{quadratic fit and validation,} \\
20^\circ : \quad & 3 \times 3 \text{ scan} \longrightarrow \text{quadratic fit and validation,} \\
10^\circ : \quad & 3 \times 3 \text{ scan} \longrightarrow \text{quadratic fit and validation,} \\
5^\circ : \quad & 3 \times 3 \text{ scan} \longrightarrow \text{quadratic fit and validation.}
\end{aligned} \tag{4.96}$$

4.5.13 Final Selection and Verification

The strongest directly measured configuration across all five stages is retained independently:

$$(\theta_{1,\text{global}}, \theta_{2,\text{global}}) = \arg \max_{(\theta_1, \theta_2) \in \bigcup_{k=1}^5 \mathcal{G}_k} C(\theta_1, \theta_2). \tag{4.97}$$

This configuration represents the largest detector count physically observed among the 45 hierarchical-search measurements.

The final model-selected angle pair is obtained from the fifth stage:

$$(\theta_{1,\text{final}}, \theta_{2,\text{final}}) = \left(\theta_{1,\text{selected}}^{(5)}, \theta_{2,\text{selected}}^{(5)} \right). \tag{4.98}$$

The polarization controller is moved to this final angle pair, and one additional valid detector reading is collected:

$$C_{\text{verify}} = C(\theta_{1,\text{final}}, \theta_{2,\text{final}}). \tag{4.99}$$

The algorithm therefore reports both the strongest directly measured configuration and the final model-selected and physically verified configuration. This distinction is necessary because the fitted model may predict an angle pair between the discrete grid points.

4.5.14 Algorithm Characteristics and Limitations

The hierarchical response-surface method reduces the number of search measurements to 45 hierarchical measurements. The final verification increases the total to 46 measurements. The approach also produces continuously valued angle predictions rather than restricting the final result to the discrete search grid. Moreover, the implementation discards clearly invalid or corrupted detector readings and repeats the measurement. However, a faulty but numerically valid measurement may still influence the fitted quadratic model or the selected search center.

In addition, the method does not guaranty discovery of the global maximum over the entire two-dimensional angle space. Especially in the early stages, it may select an incorrect region if the photon-count surface contains narrow or widely separated maxima. The method therefore assumes that high-count regions are sufficiently smooth and broad for the hierarchical measurements to identify a useful basin of attraction.

The compensation procedure assumes that the photon source, detector response, optical alignment, and environmental conditions remain approximately stable during a single search. Under this assumption, changes in the measured photon-count rate are attributed primarily to changes in the MPC paddle angles.

The fitted quadratic is a local approximation rather than a permanent physical model of the fiber link. Since the fiber transformation may vary with time, the coefficients

$$a, b, c, d, e, f \tag{4.100}$$

must be recalculated from the latest nine measurements during every stage and every compensation run. The polarization transformation should remain approximately stable throughout one search:

$$T_{\text{search}} \ll T_{\text{drift}}, \tag{4.101}$$

where T_{search} is the duration of the compensation procedure and T_{drift} is the characteristic timescale over which the fiber polarization transformation changes significantly. As a result, the hierarchical quadratic search is used as a rapid acquisition or reacquisition procedure. Its experimental performance is evaluated by comparing the obtained photon-count rate with the maximum measured using a denser reference scan. A rela-

tive performance metric can be defined as:

$$\eta = \frac{C_{\text{algorithm}}}{C_{\text{reference}}}, \quad (4.102)$$

where $C_{\text{algorithm}}$ is the count rate obtained by the hierarchical algorithm and $C_{\text{reference}}$ is the maximum count rate obtained from the dense reference measurement. The complete hierarchical polarization-compensation procedure is summarized as follows:

Algorithm 1: Hierarchical Polarization Compensation Algorithm

Input : Angle range $[0^\circ, 160^\circ]$ and step sizes $\mathcal{H} = (80^\circ, 40^\circ, 20^\circ, 10^\circ, 5^\circ)$

Output: Final angles $(\theta_{1,\text{final}}, \theta_{2,\text{final}})$ and verification count C_{verify}

$(\theta_{1,c}, \theta_{2,c}) \leftarrow (80^\circ, 80^\circ)$

$\mathcal{D} \leftarrow \emptyset$

foreach $h_k \in \mathcal{H}$ **do**

$(\theta_{1,c}, \theta_{2,c}) \leftarrow \text{Adjustcenter}(\theta_{1,c}, \theta_{2,c}, h_k)$

$\mathcal{G}_k \leftarrow \text{SnakeGrid}(\theta_{1,c}, \theta_{2,c}, h_k)$

$\mathcal{S}_k \leftarrow \text{MeasureGrid}(\mathcal{G}_k)$

$\mathcal{D} \leftarrow \mathcal{D} \cup \mathcal{S}_k$

$\hat{C}_k \leftarrow \text{FitQuadratic}(\mathcal{S}_k)$

if $\text{ValidModel}(\hat{C}_k)$ **then**

$(\theta_{1,s}, \theta_{2,s}) \leftarrow \text{PredictedMaximum}(\hat{C}_k, \theta_{1,c}, \theta_{2,c}, h_k)$

else

$(\theta_{1,s}, \theta_{2,s}) \leftarrow \arg \max_{(\theta_1, \theta_2, C) \in \mathcal{S}_k} C$

$(\theta_{1,c}, \theta_{2,c}) \leftarrow (\theta_{1,s}, \theta_{2,s})$

$(\theta_{1,g}, \theta_{2,g}, C_g) \leftarrow \arg \max_{(\theta_1, \theta_2, C) \in \mathcal{D}} C$

// Recorded for comparison

$(\theta_{1,\text{final}}, \theta_{2,\text{final}}) \leftarrow (\theta_{1,s}, \theta_{2,s})$

$\text{MoveMPC}(\theta_{1,\text{final}}, \theta_{2,\text{final}})$

$C_{\text{verify}} \leftarrow \text{ReadValidCount}()$

Chapter 5

Results and Evaluation

This chapter covers the experimental results of the automated polarization compensation system. The evaluation focuses on four main points: the photon-count rate before and after compensation, the convergence behavior of the search procedure, the time required to complete compensation, and the stability of the final compensated state.

The results are first presented using the data collected from repeated automated compensation runs. The performance of the system is then compared with other polarization compensation approaches reported in related experimental work. Finally, limitations of the present work are discussed. The purpose of this chapter is to show whether the proposed method can improve the detected photon-count rate in a repeatable and practical amount of time.

5.1 Evaluation Method

The automated polarization compensation was evaluated by running the full compensation procedure multiple times from different initial MPC paddle configurations. The positions of these paddles were randomly selected. The photon-count rate at this starting point was measured before the compensation algorithm was started. This value was used as the initial photon-count rate for that run.

Each compensation run used the same hierarchical search procedure described in [section 4.5](#). During the search, the algorithm measured the photon-count rate at different combinations of paddle-angles and selected the angle pair that produced the strongest detector response. The compensation time was measured from the start of the search

until the final compensated paddle angles were selected. The time used to move to the initial starting configuration was not included in the compensation time.

After the algorithm selected the final paddle angles, the MPC was kept fixed and the photon-count rate was measured several additional times. These repeated measurements were used to verify that the final compensated count was stable and was not caused by a single random fluctuation. The final photon-count rate was reported as the mean of these verification measurements.

The main quantities used for evaluation were the initial photon-count rate, the final verified photon-count rate, the compensation time, and the stability of the final count. The improvement in photon-count rate was calculated as:

$$\text{Improvement} = \frac{C_{\text{final}} - C_{\text{initial}}}{C_{\text{initial}}} \times 100\%, \quad (5.1)$$

where C_{initial} is the photon-count rate before compensation and C_{final} is the verified photon-count rate after compensation.

5.2 Automated Compensation Results

5.2.1 Initial and Final Photon-Count Rates

The automated compensation method was tested on ten experimental runs. Each run started from a different integer-valued MPC configuration generated using stratified sampling over the available paddle-angle range. The initial photon-count rates varied strongly between the runs, ranging from 53,312 to 823,759 cps. This shows that the compensation procedure was tested from both low-count and already high-count starting conditions.

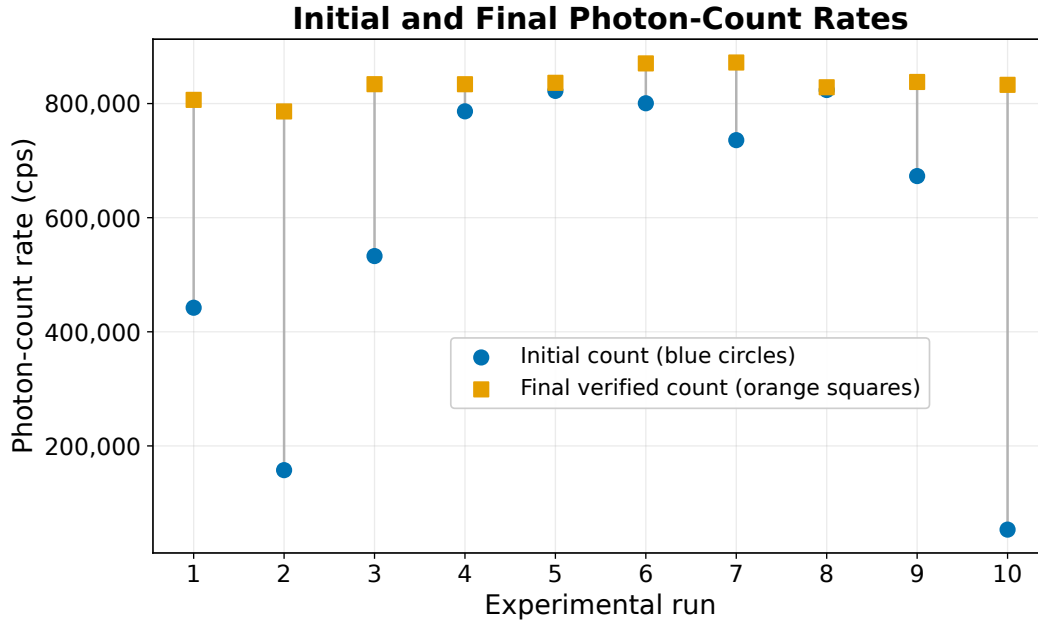


Figure 5.1: Initial and final verified photon-count rates for the ten automated compensation runs.

After compensation, the final verified photon-count rates became more consistent. The final count ranged from 786,214 cps to 871,961 cps, with a mean value of $833,864 \pm 25,512$ cps. All ten runs improved relative to their initial photon-count rate, and nine of the ten runs finished above 800,000 cps. The largest recovery occurred in Run 10, where the photon-count rate increased from 53,312 to 832,839 cps.

In addition, in most of the compensation experiments, the algorithm was able to detect the global maximum or very close neighborhood of the global maximum. For instance, figure [fig. 5.2](#) shows the selected angles from the compensation algorithm during experiment 5.

Measured Polarization Landscape

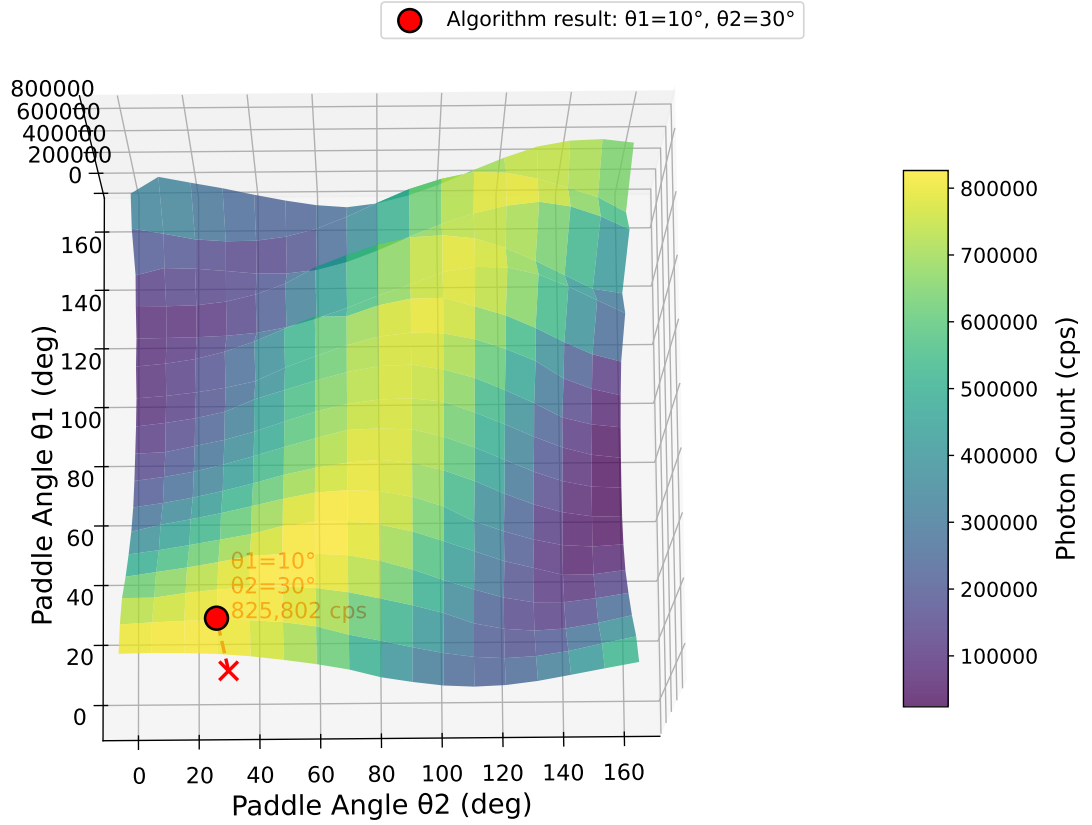


Figure 5.2: Final selected angles of Run 5.

5.2.2 Search Convergence

The convergence behavior was evaluated by tracking the highest photon-count rate found so far during each run. Across the ten runs, the search reached at least 90% of the highest measured count within the first measurement of the search. The median number of measurements required to reach 95% of the highest measured count was also one measurement, while the median number required to reach 99% was 20 measurements.

The stage-by-stage results show that the coarse search already located a high-count region in most runs. After the first stage, the best measured count was already 97.0% of the highest search count on average. This increased to 99.0% after the third stage and 99.4% after the fourth stage. The later stages therefore mainly refined the result rather than completely changing the compensation region.

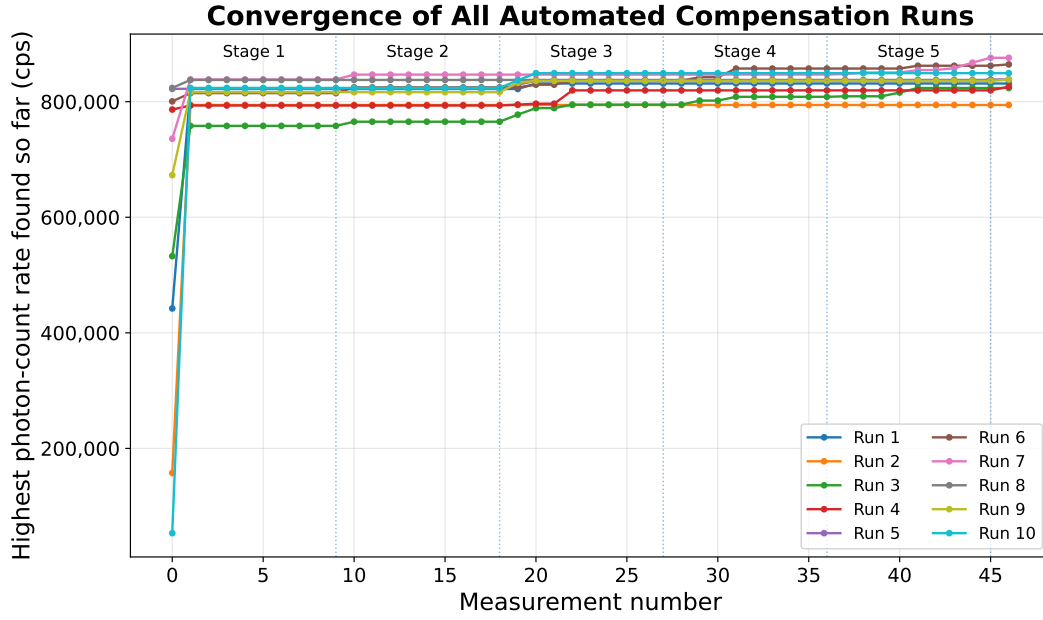


Figure 5.3: Normalized convergence of the automated compensation search across all ten runs.

As shown in [fig. 5.3](#), the search quickly identifies high-count regions during the early stages and then performs smaller refinements in the later stages. Since the plotted value is the best count found so far, each curve either increases or remains constant as the search progresses. This shows that the automated procedure consistently moves toward a strong detector response within the fixed number of measurements. For instance, [fig. 5.4](#) shows the photon count during the 5 stages of the search in experiment run-9.

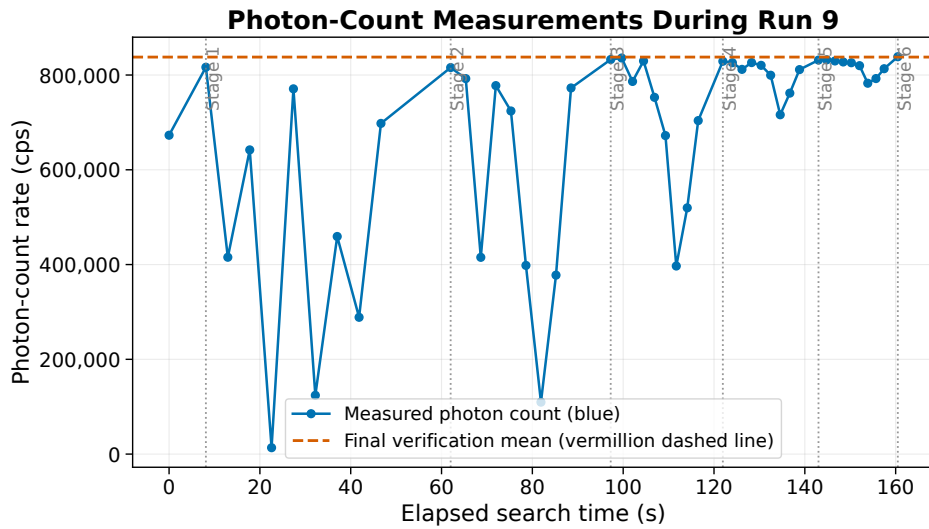


Figure 5.4: Photon-count measurements during representative compensation run 9

5.2.3 Compensation Time

The compensation time was measured from the start of the automated search until the final compensated paddle angles were selected. The time required to move to the initial starting configuration was not included. Each run used the same fixed search structure and therefore used 46 measurements.

The mean compensation time over the ten runs was 158.5 ± 4.5 s, corresponding to 2.64 ± 0.08 min. The fastest run completed in 152.4 s, while the slowest completed in 165.6 s. This shows that the automated method completed compensation consistently within three minutes.

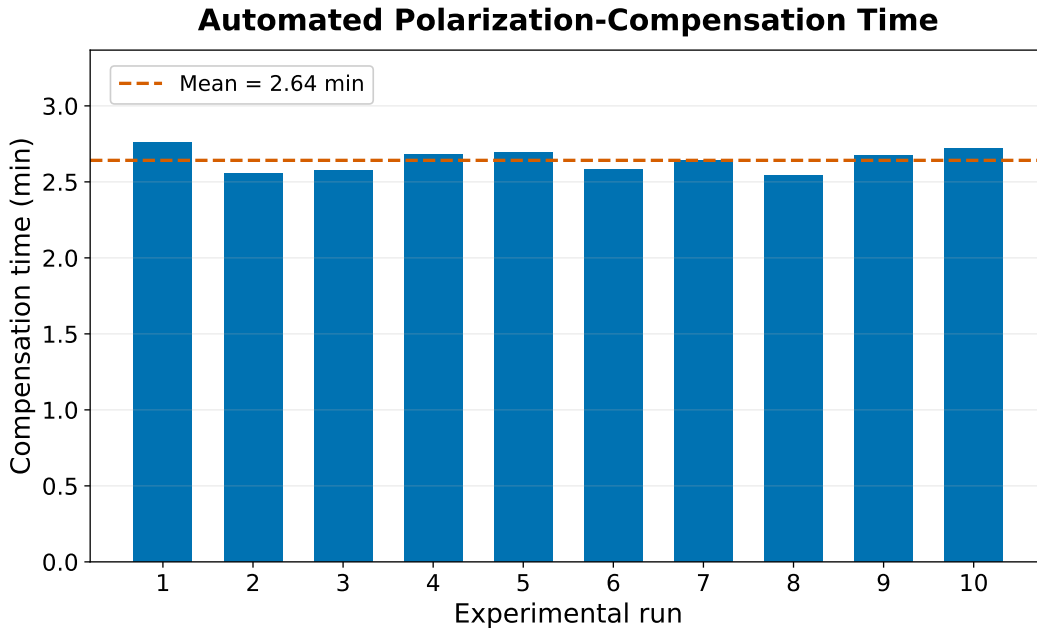


Figure 5.5: Automated compensation time for each experimental run.

5.2.4 Final Count Stability

After the search ended, the selected paddle angles were kept fixed and the photon-count rate was measured five additional times. These measurements were used to verify that the final result was stable and was not caused by a single random fluctuation.

The final verification measurements showed a small variation. The mean standard deviation of the final verification counts was 3,887 cps. Relative to the final photon-count rate, this corresponds to an average variation of 0.47%. This indicates that the compensated photon-count rate remained stable over the verification window.

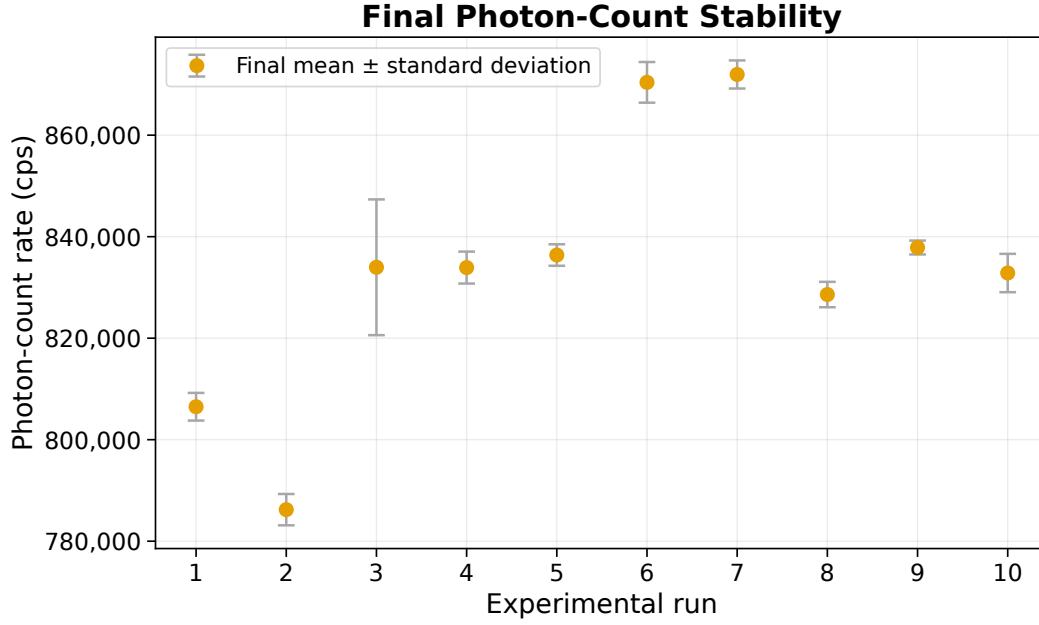


Figure 5.6: Final photon-count stability after automated compensation.

Overall, the results show that the automated MPC-based compensation method reliably recovered a high photon-count rate from different initial configurations. The final photon-count rate was consistently high across the ten runs, the compensation time remained below three minutes, and the repeated final measurements showed that the compensated state was stable over the verification period. [Table 5.1](#) summarizes the results described above.

Table 5.1: Summary of the automated compensation results over ten runs.

Metric	Result
Number of runs	10
Successful runs	10
Initial photon-count range	53,312–823,759 cps
Mean initial photon-count rate	$582,754 \pm 282,709$ cps
Final photon-count range	786,214–871,961 cps
Mean final photon-count rate	$833,864 \pm 25,512$ cps
Runs improved after compensation	10/10
Runs above 800,000 cps after compensation	9/10
Mean compensation time	158.5 ± 4.5 s
Measurements per run	46
Final verification samples per run	5
Mean final-count variation	0.47%

5.3 Comparison with Existing Compensation Methods

The following selected studies address polarization control using different feedback signals, controller technologies, and experimental objectives. Hence, their reported visibility, fidelity, QBER, photon-count rate, and compensation time values are not directly interchangeable. The comparison presented here is therefore not intended to rank the methods. Instead, it shows how the present detector-feedback approach relates to other ways of measuring and correcting polarization changes. Table 5.2 provides a summary of the polarization compensation approaches.

Table 5.2: Summary of the polarization control approaches considered in this comparison.

Work	Feedback and control	Reported outcome
Peranić et al. [35]		
<i>MPC realization</i>	Polarization visibility in the <i>HV</i> and <i>DA</i> bases. Motorized polarization controllers are adjusted sequentially using visibility thresholds.	$(98.4 \pm 0.2)\%$ visibility in approximately 8 min per link.
QUANT-NET [36]	Two classical reference states are measured using a polarimeter, and particle swarm optimization determines the controller settings.	Fidelity above 99.5% over a 4.63 km deployed fiber link.
Nakamura et al. [37]	Interference visibility is used as feedback while two polarization-maintaining fiber stretchers are adjusted.	Visibility maintained mainly above 99.9% for three hours, with 0.02 dB controller loss.
Present work	SNSPD photon-count feedback is used with two active MPC320 paddles and a hierarchical response-surface search.	Mean compensation time of 2.64 ± 0.08 min using 46 measurements.

Peranić et al. examine four polarization compensation methods [35]. However, only their motorized-polarization controller realization is considered here because it provides the closest direct comparison with the present work. Both systems automate the rotation of fiber loops to modify the stress-induced birefringence experienced by transmitted light. Nevertheless, the two implementations also differ in their feedback signals, search procedures, and experimental objectives.

In the system of Peranić et al., a tunable auxiliary laser is used to send predefined horizontal and diagonal polarization states through the fiber. When the horizontal state is transmitted, the detector corresponding to the orthogonal vertical state is monitored. Likewise, when the diagonal state is transmitted, the anti-diagonal detector is monitored. Motorized polarization controllers are adjusted to minimize these orthogonal detector counts and thus increase the polarization visibility in the HV and DA bases.

Their algorithm first determines which controller paddle has the greatest influence on the measured visibility. That paddle is adjusted until visibility increases, after which another paddle is selected when the required threshold has not yet been reached. The angular step is reduced as the measured visibility approaches the target. The procedure evaluates the two measurement bases and stops when the defined basis or global visibility threshold is satisfied. Using this procedure, the authors report a polarization visibility of $(98.4 \pm 0.2)\%$ and an average compensation time of approximately 8 min per link.

The present work also controls rotating fiber loops, but the measurement objective is different. A Thorlabs MPC320 is used with fiber installed around two of its three motorized paddles. Instead of sending predefined polarization states and calculating visibility in two bases, the present system uses the photon-count rate measured at one selected SNSPD channel after the FPBS. The two paddle angles are therefore selected according to:

$$(\theta_1^*, \theta_2^*) = \arg \max_{\theta_1, \theta_2} C(\theta_1, \theta_2), \quad (5.2)$$

where $C(\theta_1, \theta_2)$ is the measured photon-count rate.

The search strategies are also different. Peranić et al. adjust paddles according to their individual influence on polarization visibility and use threshold-based stopping conditions. In the present work, the two paddle-angles are treated as coupled variables. Five hierarchical 3×3 searches are performed using angular step sizes of 80° , 40° , 20° , 10° , and 5° . At each stage, a quadratic response surface is fitted to the nine measured counts. A validated model prediction, or the strongest directly measured point when the model is rejected, defines the next search region. By doing this, the present system completed the fixed sequence of 46 measurements in 2.64 ± 0.08 min, compared with approximately 8 min per link for the motorized realization of Peranić et al.

The final quantities reported are similarly different. Peranić et al. report polarization

visibility, which describes the contrast between the detector outcomes in the selected measurement bases. The present work reports the photon-count rate obtained at the selected SNSPD channel. Across ten runs, the present system produced a mean final count of $833,864 \pm 25,512$ cps, with an average variation of 0.47% during the five final verification measurements. This result shows that the algorithm repeatedly located a stable high-count configuration but does not independently provide a measurement of polarization visibility, QBER, or entanglement fidelity. Moreover, this shows that motorized polarization controllers can be used in different ways, which may lead to even better polarization compensation algorithms in future implementations.

The QUANT-NET system provides a comparison with polarization control on a deployed network link [36]. It transmits horizontal and diagonal classical reference states and measures the resulting Stokes parameters using a polarimeter. Particle swarm optimization (PSO) is then used to determine the three voltage settings of an electrically driven polarization controller. According to the paper, over a 4.63 km deployed fiber link, the system maintains a polarization state fidelity above 99.5% for both reference states. The calibration procedure is also integrated into the QUANT-NET control plane which allows calibration tasks to be scheduled and coordinated between network nodes.

On the other hand, the present system does not use a polarimeter or reconstruct the Stokes parameters. The search algorithm observes only the SNSPD count produced by each pair of MPC angles. This allows the optical link to be treated as a black-box response, but it also means that the final state is not identified completely. QUANT-NET demonstrates scheduled link stabilization as part of a larger network-control system, whereas the present experiment evaluates the acquisition of a high-count configuration on one selected link.

Nakamura et al. addressed polarization mismatch between two paths of a fiber interferometer [37]. Their circle-circle crosspoint method uses two polarization-maintaining fiber stretchers. The stretcher voltages are adjusted alternately using interference visibility as feedback, and the adjustment step is reduced as the visibility approaches its maximum. The experiment maintains visibility mainly above 99.9% for three hours. The measured insertion loss of each custom fiber stretcher is 0.02 dB.

The objective of that system differs from that in the present work because it matches the polarizations of two interferometer paths rather than directing photons toward one output of a polarization analyzer. It also places particular emphasis on low optical loss and continuous visibility stabilization. The present system instead uses a commercial MPC320 installed on a transmission link. Its insertion loss was not evaluated in this

experiment, and its stability measurement was limited to five verification samples after each compensation run rather than a multi-hour stabilization test.

Overall, the present work demonstrates a fast, stable, and reproducible method for automated polarization compensation using only the photon-count rate available from the SNSPD channel. It focuses directly on quantity required by the experimental link and compensates for polarization drift without the need for reconstructing the complete polarization state or introducing a separate calibration signal. Across ten independent runs, the same fixed 46-measurement procedure successfully improved the photon-count rate in all cases and completed compensation in 2.64 ± 0.08 min. The small variation in the compensation time shows that the procedure is predictable, while the mean final count of $833,864 \pm 25,512$ cps and the average final count variation of only 0.47% show that the selected configurations were strong and stable throughout the verification period.

In comparison, the motorized controller implementation reported by Peranić et al. required approximately 8 min per link, although it performed a more complete two-basis visibility calibration and hence addressed a different objective. Therefore, the strength of the present method is its ability to combine rapid acquisition, repeatable operation, and direct single-photon feedback in a compact control procedure. This makes it a practical foundation for systems in which polarization must be restored automatically after a link is established, disturbed, or reconfigured.

5.4 Limitations of the Present Work

Although the experimental results demonstrate fast and reproducible polarization compensation, the evaluation was performed on one selected optical link using a fixed laboratory configuration. Moreover, the ten experimental runs confirm that the method repeatedly recovers a high photon-count rate under the tested conditions. However, the same performance has not yet been verified across different fiber lengths, optical paths, detector channels, or multiple network nodes.

The optimization objective is also limited to the photon-count rate measured at one selected SNSPD channel after the FPBS. A high count indicates that a large fraction of the received photons is directed toward the selected polarization output. However, this measurement does not reconstruct the complete polarization state and does not independently determine polarization visibility, QBER, or entanglement fidelity. Changes in source brightness, optical loss, coupling efficiency, or detector response could also affect

the measured count rate. The current method therefore assumes that these quantities remain approximately stable during one compensation run.

The hierarchical search also does not guarantee that the global maximum of the complete paddle-angle landscape will always be found. The method assumes that a useful high-count region is sufficiently broad and smooth to be identified during the early search stages. A narrow maximum located between the coarse initial measurements could be missed. The validated quadratic model and measured-point fallback reduce the effect of unreliable local predictions, but they do not provide a mathematical guarantee of global optimality.

The reported stability was evaluated using five verification measurements collected after each compensation run. These measurements show that the selected count rate remained stable over the short verification interval, with an average variation of 0.47%. However, the experiment did not evaluate stability over several hours or days, nor did it continuously compensate for polarization drift while the link was operating. Recovery from deliberately introduced disturbances was also not evaluated.

Finally, the present system performs the 46 detector measurements sequentially on a single link. Its operation has not yet been integrated with network routing, calibration scheduling, or simultaneous compensation of multiple optical links. Therefore, the results demonstrate rapid and repeatable acquisition of a high-count polarization configuration under the tested laboratory conditions, rather than continuous network-wide polarization stabilization.

Chapter 6

Conclusion

The motivation for this thesis begins with the difficulty of scaling a single quantum processor. Distributed quantum computing provides another path by connecting smaller quantum systems through optical links and allowing them to share quantum information and entanglement. However, establishing a route between two quantum-network nodes is not sufficient on its own. The selected physical path must also deliver photons under the optical conditions required by the receiver.

Polarization is one of these conditions. As photons propagate through the optical fiber, bending, mechanical stress, temperature changes, and other environmental effects can transform their polarization state. A link that has been successfully established at the network level may therefore still be unusable at the physical level if the received polarization is not correctly aligned. In small laboratory experiments, this problem can be handled by manually adjusting the polarization controllers. However, such an approach depends on the operator's skills and becomes more difficult to repeat as the number of links and possible optical paths increases.

Thus, this thesis approached polarization compensation as an automation problem. The detector, polarization controller, and control computer were combined into a closed feedback loop, which removed the need for an operator to observe the detector output and manually rotate the controller paddles. The SNSPD provides information about the received optical signal, the compensation algorithm interprets this information, and the motorized polarization controller applies the corresponding physical adjustment. In this way, the detector output serves both as the final result of the optical experiment and as feedback for configuring the optical link.

The photon-count rate was selected as the feedback quantity because it is already avail-

able on the receiving side of the experimental platform and directly reflects how much of the received signal is directed toward the selected FPBS output. This avoids requiring a complete analytical model of the fiber transformation during the compensation process. Instead, the optical link is treated as an unknown response that can be measured and improved experimentally. This choice keeps the control path closely connected to the equipment already used by the quantum link.

The relationship between the two MPC paddle angles and the measured photon-count rate is coupled, nonlinear, and not known in advance. In addition, every additional measurement requires physical movement of the paddles, settling time, and detector acquisition. An exhaustive search would therefore provide broad coverage, but would introduce an impractical delay. Hence, the developed method uses a hierarchical search that first identifies a useful region of the two-dimensional angle space and then progressively refines it. The response-surface model uses the measurements collected at each stage to guide the next stage, while the measured-point fallback prevents an unreliable model from controlling the search.

Using this approach, the thesis connects the original network-level motivation to a practical physical-layer implementation. It shows how a motorized polarization controller, a single-photon detector, and a software-based search procedure can be combined to replace a manual alignment task with a repeatable automated process. The main contribution is therefore not only the selection of a particular pair of paddle angles, but also the development of a complete control path through which an optical link can measure its own condition and adjust itself with minimal human intervention.

This is an important distinction for future quantum networks. As optical paths are established, changed, or disturbed, their physical configuration must also be prepared for quantum communication. Polarization compensation can therefore be considered to be part of the link configuration rather than a one-time laboratory alignment procedure. The system developed in this thesis provides an initial implementation of that idea: *Adaptive Polarization Compensation for Optical Interconnects in Quantum Networks*.

Future Work

The first immediate extension of this work is to transform the current compensation procedure into a continuously operating polarization monitor. The algorithm has been successfully tested as a laboratory calibration procedure, but it has not yet been integrated as a background process that continuously checks the condition of the quantum

link. In a future implementation, the Raspberry Pi could monitor the photon-count rate for as long as the link remains established and automatically trigger the polarization-compensation algorithm when a significant reduction is detected.

To support this process, an expected photon-count rate could first be determined during link initialization. This expected value should account for the photon-source rate, optical transmission loss, detector efficiency, and the count normally directed toward the selected FPBS output. The monitoring process could then compare the measured count rate with a threshold defined as:

$$C_{\text{th}} = \alpha C_{\text{expected}}, \quad 0 < \alpha < 1, \quad (6.1)$$

where C_{expected} is the calibrated count rate and α determines the permitted reduction before compensation is required. If the measured count remains below C_{th} for several consecutive acquisition windows, the system could treat this as a possible polarization change and automatically begin compensation.

Moreover, such a monitoring process would change the role of the present algorithm. Instead of being started manually whenever compensation is required, it would become part of the operation of the quantum link. The system would first compensate for the link after it has been established, and then continue to monitor the detector response. If bending, temperature changes, mechanical disturbance, or path reconfiguration cause the count rate to fall, the link could temporarily enter a calibration state, recover a high-count MPC configuration, and then return to normal operation.

The second important direction is to test the method using deployed optical fiber outside the controlled laboratory environment. The deployed fiber is subject to larger and less predictable changes caused by environmental temperature, vibration, mechanical loading, and daily operating conditions. Testing the compensation system over those links would show whether the polarization compensation algorithm remains effective when the polarization transformation changes under realistic conditions.

This deployed-fiber evaluation could be performed as part of an entanglement distribution or entanglement-swapping experiment. In such a setup, polarization compensation would be performed on the fiber links connecting the photon source, remote nodes, and Bell-state analyzers. The algorithm could then be evaluated according to its effect on the complete quantum experiment rather than only on the photon count at one detector output. For an entanglement-swapping experiment, useful measurements could include the Bell-state consumption rate, coincidence rate, interference visibility, and the fidelity

of the entangled state produced between the remote nodes.

Another extension is to evaluate the response of the system to controlled disturbances. The fiber could be deliberately bent, heated, cooled, or mechanically moved while the monitoring process is active. The time between the disturbance and its detection, the time required for compensation, and the quality of the recovered state could be measured separately. This would provide a clearer evaluation of the complete sequence:

$$\text{disturbance} \rightarrow \text{detection} \rightarrow \text{re-compensation} \rightarrow \text{recovery}. \quad (6.2)$$

The compensation algorithm could also be extended to use information from both FPBS detector channels. A single-channel objective can increase the count at the selected output, but a two-channel metric could distinguish between an actual polarization improvement and a general change in source brightness or optical transmission. For example, the normalized quantity:

$$M = \frac{C_{\text{target}}}{C_{\text{target}} + C_{\text{orthogonal}}} \quad (6.3)$$

could be used as an optimization objective. Because this metric depends on the distribution of photons between the two polarization outputs, it would be less sensitive to variations that affect both detector channels at the same time.

Finally, the system should be extended from one link to multiple reconfigurable optical paths. Each path may produce a different polarization transformation and may therefore require a different MPC configuration. The network controller could store previously obtained settings for each path and use them as starting points when the same path is selected again. The compensation procedure could then be automatically activated after routing, switching, or link reconfiguration. This would connect the physical-layer compensation method developed in this thesis with higher-level network control and move the system closer to autonomous operation in a multi-node quantum network.

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